QAL TECHNICAL SPECIFICATION & ARCHITECTURAL DOSSIER

Out-of-Band Acoustic Physical-Presence Immobilizer & Multi-Factor Automotive Authentication System

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Core Regulatory & Engineering Compliance Baseline

This specification and demonstration architecture are engineered to align with global automotive cybersecurity, functional safety, and cryptographic principles:

Regulatory Standard	Scope & Domain within QAL Architecture
ISO/SAE 21434:2021	Road Vehicles - Cybersecurity Engineering (Threat modeling, attack mitigation, and cryptographic hardware binding).
UNECE WP.29 R155 / R156	Cyber Security and Software Update Management (Immutable hash-chained forensic blackbox logging and incident auditability).
ISO 14229-1 (UDS)	Unified Diagnostic Services (Service 0x27 SecurityAccess challenge-response and diagnostic parameter DIDs 0xF180-0xF185).
CCC Digital Key Release 3	Architectural Interoperability Target (Conceptual alignment with Car Connectivity Consortium key-sharing and role-delegation models).
ISO 26262 (ASIL-B/D Target)	Functional Safety Target (Designed to interface with ASIL-rated powertrain interlocks, high-voltage contactors, and starter motor relays).
UK / EU GDPR (Art. 5 & 17)	Privacy & Data Protection (Zero plaintext storage on non-volatile memory, bounded circular audit buffers, and statutory ownership de-provisioning).

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Master Nomenclature & Acronyms Glossary

The following standardized automotive, cryptographic, and signal-processing definitions govern the architectural specification and implementation of the Quackworks Acoustic Lock (QAL) system:

Acronym / Term	Domain	Formal Definition & Engineering Context within QAL
2FA	Security	Two-Factor Authentication: Multi-factor gating requiring both primary wireless ingress (Factor 1) and in-cabin acoustic verification (Factor 2) before releasing propulsion interlocks.
ADC	Hardware	Analog-to-Digital Converter: Hardware digitizing analog cabin telematics microphone signals at 44.1 kHz or 48.0 kHz.
AES-256-CBC	Cryptography	Advanced Encryption Standard (256-bit, Cipher Block Chaining): Symmetric cipher encrypting the permanent Vehicle Secret within the non-volatile vault.
AFP	Diagnostics	Authorised Forensic Panel: Statutory diagnostic terminal utilized by accident reconstruction and vehicle recovery units to extract blackbox telemetry and release Shift-by-Wire neutral locks.
AM	Signal Processing	Amplitude Modulation: Undesirable volume-fluctuation sidebands eliminated by replacing per-bit Hann windowing with single-burst raised-cosine fades.
AMR / AMR-WB	Telecommunications	Adaptive Multi-Rate (Wideband) Audio Codec: Cellular voice compression standard cutting off audio above 3.4-7.0 kHz, defeating remote cellular audio streaming attacks.
ANC	Acoustics	Active Noise Cancellation: Factory vehicle headliner microphones capable of capturing 16.0-17.0 kHz near-ultrasonic tokens.
APP	Powertrain	Accelerator Pedal Position: Drive-by-wire sensor voltage line clamped to zero-torque by QAL immobilizer gateway interlocks.
ASIL	Safety	Automotive Safety Integrity Level: Risk classification under ISO 26262. (QAL targets ASIL-B/D functional safety interfaces).
ATP	Diagnostics	Authorised Technician Panel: Workshop maintenance gateway conforming to ISO 14229 UDS Service 0x27 for servicing and clearing anti-tamper lockouts.
AUTOSAR	Architecture	Automotive Open System Architecture: Standardized software framework for vehicle compute modules, crypto stacks, and SecOC.
BCM	Hardware	Body Control Module: Electronic compute unit managing vehicle access, door latches, exterior lighting, and keyless wake-up triggers.
BER	Telecommunications	Bit Error Rate: Percentage of corrupted bits during acoustic demodulation (0% in-cabin target).
BEV / PHEV / FCEV	Powertrain	Battery / Plug-in Hybrid / Fuel Cell Electric Vehicle: Powertrain architectures whose high-voltage battery contactors and inverter gate drivers are gated by QAL.

Acronym / Term	Domain	Formal Definition & Engineering Context within QAL
BLE	Wireless	Bluetooth Low Energy (2.4 GHz): Short-range wireless standard used for primary ingress keys and offline mobile pairing.
BOM	Manufacturing	Bill-of-Materials: Physical hardware and manufacturing component cost ($0 added vehicle hardware for QAL).
CAN / CAN-FD	Networking	Controller Area Network (Flexible Data-Rate): Primary vehicle communication bus interconnecting the ECU, BCM, and gateway modules.
CCC	Standards	Car Connectivity Consortium: Global cross-industry organization standardizing smartphone-to-vehicle digital key access (Digital Key Release 3).
CP-FSK	Modulation	Continuous-Phase Frequency-Shift Keying: Orthogonal frequency modulation maintaining continuous phase across symbol transitions to prevent wideband splatter.
CRC32	Integrity	Cyclic Redundancy Check (32-bit, IEEE 802.3): Polynomial checksum (0xEDB88320) verifying acoustic bitstream transmission integrity.
DAC	Hardware	Digital-to-Analog Converter: Sound generation chip in user smartphones and vehicle infotainment systems.
DID	Diagnostics	Data Identifier: Diagnostic parameter registers defined under ISO 14229 (e.g., 0xF180 Hardware Version, 0xF183 Calibration ID).
DoIP	Networking	Diagnostics over Internet Protocol (ISO 13400): High-speed diagnostic transport over automotive Ethernet.
DSP	Processing	Digital Signal Processing: Real-time algorithmic audio filtering, I/Q demodulation, and bandpass isolation.
DTC	Diagnostics	Diagnostic Trouble Code: Standardized hexadecimal error codes (e.g. 0x00 to 0x0A) logged during acoustic validation.
ECDH	Cryptography	Elliptic-Curve Diffie-Hellman: Asymmetric key-agreement protocol (P-256 curve) securing mobile device pairing.
ECM	Powertrain	Engine Control Module: Compute unit controlling internal combustion ignition timing and fuel injector pulses.
ECU	Hardware	Electronic Control Unit: Embedded automotive computer executing QAL state machine logic and cryptographic verification.
EPB	Chassis	Electronic Parking Brake: Motorized caliper clamping system utilized as a secondary immobilization anchor.
EPDM	Materials	Ethylene Propylene Diene Monomer: High-density synthetic elastomeric rubber used in vehicle door weatherstripping providing acoustic boundary attenuation.
ESCL	Hardware	Electronic Steering Column Lock: Motorized deadbolt mechanism locking the steering wheel column.
EVITA	Cybersecurity	E-safety Vehicle Intrusion Protected Applications: European automotive hardware security standard defining Light, Medium, and Full HSM architectures.

Acronym / Term	Domain	Formal Definition & Engineering Context within QAL
FFT	Signal Processing	Fast Fourier Transform: Algorithmic conversion of time-domain audio samples into discrete frequency spectrum bins.
FIFO	Data Management	First-In, First-Out: Circular ring buffer data structure capped at 100 entries conforming to GDPR storage limitations.
GDPR	Legal / Privacy	General Data Protection Regulation (UK/EU): Data privacy framework mandating data minimization (Art. 5) and right to erasure (Art. 17) during ownership transfer.
GPS	Telemetry	Global Positioning System: Geographic coordinates recorded to tamper logs during security state transitions.
HMI	Software	Human-Machine Interface: In-vehicle touchscreen console and mobile app graphical user interfaces.
HMAC-SHA256	Cryptography	Hash-Based Message Authentication Code (SHA-256): 256-bit cryptographic digest verifying token authenticity and vehicle secret binding.
HSM / SHE	Hardware	Hardware Security Module / Secure Hardware Extension: Isolated silicon enclave storing root vehicle keys and executing cryptographic hashes.
HV	Powertrain	High-Voltage: EV traction battery systems (≥ 400 V) whose main DC contactors and pre-charge relays are gated by QAL.
I/Q	Modulation	In-Phase and Quadrature: Non-coherent energy detection calculating energy values ($M = I^2 + Q^2$) regardless of wave phase angle.
ICE	Powertrain	Internal Combustion Engine: Traditional petrol/diesel powertrains whose starter relays and fuel pumps are gated by QAL.
IEEE	Standards	Institute of Electrical and Electronics Engineers: Standards body governing CRC32 (802.3) and UWB (802.15.4z).
IGBT / SiC MOSFET	Electronics	Insulated-Gate Bipolar Transistor / Silicon Carbide MOSFET: Power inverter switching stages disabled by QAL propulsion interlocks.
ISO	Standards	International Organization for Standardization: Standardizing ISO 21434 (Cybersecurity), ISO 14229 (UDS), and ISO 26262 (Functional Safety).
IV	Cryptography	Initialization Vector: 16-byte random salt ensuring identical passwords produce distinct AES ciphertexts.
LIN	Networking	Local Interconnect Network: Low-cost single-wire automotive communication bus.
LPF	Signal Processing	Low-Pass Filter: 12th-order Butterworth digital filter attenuating audible chime harmonics > 80 dB at 16.0 kHz.
MRC	Signal Processing	Maximal Ratio Combining: Diversity combining technique summing weighted SNR across multiple cabin microphones.
NFC	Wireless	Near-Field Communication (13.56 MHz): Contactless smartcard access protocol used as Factor 1 ingress.

Acronym / Term	Domain	Formal Definition & Engineering Context within QAL
NIST	Standards	National Institute of Standards and Technology: Governing PBKDF2 iteration guidelines (SP 800-132) and AES key wrapping.
NVM	Hardware	Non-Volatile Memory: On-chip EEPROM or flash memory preserving hash-chained forensic audit logs through power loss.
OBD-II	Diagnostics	On-Board Diagnostics II: Mandatory 16-pin diagnostic port under the steering column protected by QAL's independent propulsion firewall.
OEM	Industry	Original Equipment Manufacturer: Commercial vehicle manufacturer (e.g., JLR, BMW, Ford).
PBKDF2	Cryptography	Password-Based Key Derivation Function 2: Algorithm executing 100,000 HMAC-SHA256 iterations to transform user passwords into 256-bit AES keys.
PII	Legal / Privacy	Personally Identifiable Information: Protected user data (names, precise GPS, device IDs) pseudonymized in QAL audit logs.
PKES	Access Control	Passive Keyless Entry and Start: Vulnerable radio-frequency key fob access system reinforced by QAL's out-of-band acoustic layer.
PKI	Cryptography	Public Key Infrastructure: Asymmetric certificate framework governing ISO 14229 UDS 0x27 diagnostic sessions.
PVB	Materials	Polyvinyl Butyral: Viscoelastic acoustic damping interlayer laminated inside automotive safety glass providing high-frequency transmission loss.
PWM	Control	Pulse-Width Modulation: High-frequency gate signals controlling electric motor inverters.
QAL	Architecture	Quackworks Acoustic Lock: The present out-of-band near-ultrasonic physical-presence vehicle authentication and immobilizer system.
RBAC	Security	Role-Based Access Control: Security architecture enforcing strict privilege separation between Master Owner, Delegated Guests, and Technicians.
RSA	Threat Model	Relay Station Attack: Cyber-physical vehicle theft utilizing wireless amplifier bridges to spoof key fob proximity.
RTC	Hardware	Real-Time Clock: Internal battery-backed hardware timer providing monotonic UTC timestamps for audit logging and time-decay evaluation.
SAE	Standards	Society of Automotive Engineers: Professional standards body governing ISO/SAE 21434 and SAE J2012 diagnostic trouble codes.
SbW	Powertrain	Shift-by-Wire: Electronic transmission park-lock actuator released by QAL upon Stage 2 authorization or police warrant token.

Acronym / Term	Domain	Formal Definition & Engineering Context within QAL
SecOC	Architecture	Secure Onboard Communication: AUTOSAR standard verifying message freshness and cryptographic MACs across CAN buses.
SHA-256	Cryptography	Secure Hash Algorithm (256-bit): One-way cryptographic hash function used in hash-chained audit logging and pairing certificates.
SHE	Cybersecurity	Secure Hardware Extension: On-chip hardware security module specification for automotive microcontrollers.
SNR	Signal Processing	Signal-to-Noise Ratio (dB): Ratio of acoustic signal power to ambient environmental background noise.
SOF	Telecommunications	Start-of-Frame: Predefined 16-bit binary delimiter (1110001110001110) indicating the start of the cryptographic payload.
SOS	Signal Processing	Second-Order Section: Biquad filter building block cascaded six times to construct the 12th-order Butterworth low-pass filter.
SPL	Acoustics	Sound Pressure Level (dB SPL): Physical logarithmic measure of acoustic sound pressure relative to 20 micropascals.
STRIDE	Cybersecurity	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege: Threat classification model (under ISO/SAE 21434).
TARA	Cybersecurity	Threat Assessment and Remediation Analysis: Formal risk assessment methodology defined under ISO/SAE 21434.
TCU	Telematics	Telematics Control Unit: Vehicle modem providing GPS location data and secure cloud synchronization.
TL	Acoustics	Transmission Loss (dB): Structural acoustic damping and boundary attenuation provided by automotive glass and door seals (≥ 30 dB).
ToF	Physics / RF	Time-of-Flight: Wireless or acoustic distance measurement calculated from signal propagation delay.
UDS	Diagnostics	Unified Diagnostic Services (ISO 14229): Standardized diagnostic protocol used across automotive ECUs for diagnostics, flash updates, and security access.
UNECE	Regulation	United Nations Economic Commission for Europe: Regulatory body mandating vehicle cybersecurity management (WP.29 R155/R156).
UTC	Standard Time	Coordinated Universal Time: Standard astronomical timestamp reference used across all QAL token expirations and audit logs.
UWB	Wireless	Ultra-Wideband (IEEE 802.15.4z): High-frequency RF pulse ranging technology compared against QAL in Section 2.2.
VIN	Identification	Vehicle Identification Number: Unique 17-character alphanumeric identifier cryptographically bound to the ECU root certificate.
VoIP	Telecommunications	Voice over Internet Protocol: Digital voice streaming over data networks rejected by QAL latency and codec limits.

SECTION 1: EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

1.1 Executive Summary

Modern motor vehicles rely heavily on Passive Keyless Entry and Start (PKES) systems communicating over radio frequencies (typically 315 MHz, 433 MHz, or 2.4 GHz Bluetooth Low Energy). While convenient, these systems suffer from two fatal physical-layer vulnerabilities:

1. **Wireless Relay Station Attacks (RSA):** Radio signals easily penetrate residential walls, window glass, and vehicle body panels. Thieves use inexpensive wireless bridge amplifiers to extend the key's range from inside a house to the driveway, stealing the vehicle in seconds.
2. **Simultaneous Key and Phone Theft:** In real-world crimes (such as gym locker break-ins, luggage theft, hotel room burglaries, or bag snatches), a thief often steals both the physical car key and the owner's smartphone together. Under conventional digital key or passive keyless systems, having both items in proximity grants unrestricted engine ignition.

The Quackworks Acoustic Lock (QAL) system establishes an out-of-band physical-presence authentication layer designed to defeat both attack vectors. Instead of relying on passive, invisible automation, QAL introduces a deliberate, conscious user action: the operator must manually initiate the transmission of a near-ultrasonic acoustic token (16.0 kHz to 17.0 kHz Continuous-Phase Frequency-Shift Keying) from their authenticated mobile device.

Because acoustic energy is physically contained and attenuated by the vehicle cabin boundary (laminated safety glass, body panels, and elastomeric weatherstripping), external receivers cannot intercept an operable signal. Furthermore, because transmission requires the user to unlock their personal device (via FaceID, fingerprint, or passcode) and manually press transmit, physical possession of the car key and the phone is completely useless to a thief or any unauthorized operator.

1.2 Two-Factor Authentication (2FA) & Proof-of-Intent Model

QAL does not replace the vehicle key; it operates as an independent, secondary authorization gate:

- **Factor 1: Primary Wireless Ingress (Access Authorization)**
  The vehicle operator approaches carrying the primary key (RF key fob, BLE digital key, NFC card, or UWB transceiver). This primary credential authorizes exterior door unlocking and transitions the vehicle Electronic Control Unit (ECU) from deep sleep into an energized Stage 1 Physical-Presence Listening State. High-voltage contactors, fuel pumps, and starter motor relays remain strictly disabled.
- **Factor 2: Out-of-Band Manual Acoustic Token (Proof-of-Intent Propulsion Gating)**
  Once seated inside the cabin, the primary key alone cannot start the engine. The driver must unlock their mobile device and manually initiate transmission of the acoustic authentication token. The token is emitted near-ultrasonically into the cabin, captured by factory-installed interior microphones, demodulated, and cryptographically verified against a vehicle secret stored within the ECU Hardware Security Module (HSM). Only upon successful verification does the ECU transition to Stage 2 Authorized, closing propulsion circuits and permitting drive-away.

QAL INTENTIONAL DUAL-FACTOR FLOW

[DRIVER APPROACHES VEHICLE]	–> Doors Unlock
Factor 1: Primary Key Detected	–> ECU Wakes
(RF / BLE / UWB)	–> Stage 1 Listening (30-45s)
	–> **Powertrain Remains LOCKED**
[DRIVER INSIDE CABIN]	–> Driver unlocks phone (FaceID / Biometric / PIN)
Factor 2: Conscious Manual Action	–> Driver taps [Transmit]
(Proof-of-Intent)	–> 16.0-17.0 kHz Token Emitted
	–> Trapped by Cabin Boundary
	–> In-Cabin Mic Captures Audio
	–> Demodulated & HMAC Verified
	–> **STAGE 2: PROPULSION ENABLED**

1.3 Comprehensive Engineering Trade-Off Analysis (Pros & Cons)

Comprehensive Advantages (System Pros)

1. **Seamless Integration Alongside Existing OEM Architectures (Non-Disruptive Add-On):**
   QAL is designed not to require vehicle manufacturers to discard or redesign their existing Passive Keyless Entry and Start (PKES) systems, Body Control Modules (BCM), or RF key fobs. It functions as a modular, additive security firewall operating in parallel, gating the final powertrain enable line (starter relays, inverter gate drivers, or high-voltage contactors) without altering exterior vehicle ingress mechanisms.

2. **Robust RF Relay Mitigation & Mode-Dependent Replay Protection:**
   Conventional radio-frequency booster attacks rely on RF signals passing through residential walls and vehicle body panels. Near-ultrasound (16.0 kHz to 17.0 kHz) is structurally attenuated by automotive laminated glass and elastomeric door seals, restricting external relay bridges from capturing an operable signal-to-noise ratio. To defend against token forgery and acoustic replay:
   - **Primary Cryptographic Challenge-Response (Patent Claim 18):** In the primary operational architecture, the vehicle and user device execute an ephemeral challenge-response handshake bound to a 16-byte random challenge seed and an incremental monotonic freshness counter. Old audio transmissions cannot be replayed, because the vehicle rejects tokens carrying expired freshness values or invalid challenge seeds.
   - **Prototype Direct-Modulation Fallback:** For offline demonstration and simplified evaluation, the prototype retains a direct-modulated acoustic key path. Production deployments utilize the full freshness-bounded challenge-response mode to eliminate acoustic replay vulnerabilities.

3. **Defeating the "Gym Locker / Stolen Bag" Threat (Proof-of-Intent):**
   Mass-market automotive systems prioritize zero-friction convenience, which introduces a major security exposure: if a thief steals a gym bag or coat containing both the physical car key and the owner's smartphone, a standard keyless car will start immediately. QAL eliminates this vulnerability through the principle of Intentional Affirmation. Even if an unauthorized party holds both the physical key and the phone, they cannot transmit the acoustic token without first unlocking the mobile device's biometric security enclave (FaceID, fingerprint, or lockscreen passcode) and manually pressing transmit.

4. **Complete Offline Autonomy (Zero Cellular or Wi-Fi Dependency):**
   Vehicle security operates completely independently of cloud connectivity. QAL functions locally in subterranean parking garages, multi-story concrete structures, rural environments, or shipping containers with zero cellular reception. Cryptographic validation and time-epoch verification are executed locally between the user device and the vehicle ECU. Initial mobile device pairing can be executed offline via local Bluetooth Low Energy (BLE) or direct in-car dashboard entry.

5. **Multi-Tier Role-Based Access Control (RBAC) & Fleet Management:**
   The platform divides vehicle operation into strict, non-escalating operational roles:
   - **Master Owner:** Full administrative authority, credential vault derivation, guest key creation, and revocation.
   - **Delegated Guest Drivers (Up to 8 Slots):** Time-decayed passes (e.g., 24 hours, 7 days, 30 days) granting propulsion authorization only, with an absolute software lockout preventing guest access to owner settings or private vaults.
   - **Instant Revocation:** The owner can revoke any guest pass instantly from their mobile device or the in-car console without affecting their own master key.
   - **Dealership & Test-Drive Showroom Mode:** Dedicated station-gated profiles allowing supervised customer demonstrations and test drives without exposing permanent customer secrets.
   - **Workshop Technician Mode (ISO 14229 UDS 0x27):** Secure diagnostic maintenance access for repairers and service centers.
   - **Statutory Authority & Crash Investigation Gateway (Authorised Forensic Panel - AFP):** Gated behind verified statutory warrant tokens (e.g., POLICE-FORENSIC or WARRANT-99 in demonstration builds). Provides certified, read-only extraction of blackbox telemetry, ECU flash counters, and emergency vehicle recovery actuators (such as the Shift-by-Wire neutral park-lock release for flatbed towing).

6. **Tamper-Resistant Forensic Telemetry & Insurance Non-Repudiation:**
   Every lifecycle event is recorded in non-volatile flash memory within a SHA-256 hash-chained circular ring buffer. The ECU logs event classifications (UserUnlock, FailedAttempt, ServiceMode, LockoutTriggered, BypassEngaged), high-resolution UTC timestamps, GPS coordinates, and the exact credential fingerprint used. If an owner or criminal disputes vehicle state during an insurance claim, the blackbox log provides insurers and forensic

examiners with immutable proof of whether the vehicle was started via the master token, guest key, manual fallback PIN, or service override.

7. **User-Controlled Master Bypass (Logged Liability Toggle):**
   If an owner finds secondary authentication inconvenient during specific activities (e.g., track days, private property use, or continuous multi-stop deliveries), the system can be toggled to a disabled bypass state directly from Owner Mode. To eliminate legal and insurance liability ambiguity, engaging or disengaging the bypass writes a permanent cryptographic entry to the forensic log, providing complete legal non-repudiation.

8. **Hardware Form-Factor Versatility (Wearables & Smart Accessories):**
   While the smartphone application represents the primary interface, the acoustic transmission pipeline (16.0 kHz to 17.0 kHz CP-FSK) can be executed by any miniature smart device equipped with a basic audio digital-to-analog converter (DAC) and transducer. As disclosed in Patent Specification Paragraph [0128], the acoustic transmitter can be integrated into smartwatches, fitness bands, smart rings, key pendants, or specialized automotive smart accessories.

9. **Low Incremental Hardware Cost & Resource Re-Use:**
   Unlike Ultra-Wideband (UWB IEEE 802.15.4z) systems that add significant costs in specialized radar transceivers, dedicated high-frequency cabling, and multiple antennas distributed across vehicle body panels, QAL is designed to avoid requiring dedicated vehicle hardware where suitable existing vehicle microphones and processing infrastructure are available:
   - **Emission (Client-Side Masking):** Handled entirely by the driver's existing smartphone speaker, smartwatch, or wearable transducer. The user's device synthesizes the complete composite audio burst, mixing the audible masking chime (frequencies below 14.0 kHz) directly with the near-ultrasonic data payload (16.0 kHz to 17.0 kHz) into a single transmission.
   - **Capture:** Designed to utilize pre-existing, factory-installed hands-free telematics microphones already fitted to the vehicle for voice commands and phone calls, subject to vehicle-specific acoustic bandwidth validation.
   - **Zero Vehicle Speaker Requirement:** The vehicle itself is not required to emit any sound to complete authentication. Factory in-cabin vehicle speakers may optionally be utilized in advanced OEM configurations to emit an auxiliary challenge sequence, but the core immobilizer operates completely independently of vehicle audio playback systems.

10. **Elimination of Dashboard Display "Smudge Attacks":**
    Standard touchscreen PIN-to-drive systems suffer from a physical security exposure where drivers repeatedly tap the exact same numerical digits on the central infotainment display, leaving oily fingerprint smudges on the glass. QAL eliminates routine touchscreen interaction during normal startup workflows.

Engineering Challenges, System Limitations & Built-In Mitigations (Cons & Mitigations)

1. **Limitation: Fallback Vulnerability to Compromised Credentials + Stolen Key**
   The Challenge: If an unauthorized party physically steals the authentic vehicle key and also discovers or coerces the owner's private text password, they could bypass the acoustic channel by entering the manual fallback PIN directly on the in-cabin screen.
   The Mitigation: While acknowledged, this architecture remains significantly more secure than standard OEM vehicles on the road today:
   - **Standard OEM Car:** Possession of the physical key alone grants immediate engine starting.
   - **QAL Architecture:** Even with the physical key, an unauthorized party cannot start the car unless they also possess the secret knowledge factor (the owner password).
   - **State-Gated Temporal Penalty:** The manual fallback interface does not appear immediately upon sitting in the car; it remains dormant and unpowered until the primary 30-to-45 second Stage 1 acoustic window times out.
   - **Configurable Progressive Anti-Brute-Force Lockout:** Per Patent Claims 14 and 24, the ECU enforces a predetermined cooldown interval upon three consecutive incorrect attempts within a rolling window. Production implementations utilize a progressive backoff curve (30s -> 1m -> 2m -> 3m -> 5m), preventing driver lockouts caused by minor typing errors while rendering automated mathematical brute-force attacks unviable.

2. **Limitation: Intentional Friction (The 2-to-3 Second User Action)**
   The Challenge: Requiring a driver to unlock their phone (FaceID/PIN) and tap transmit breaks the completely passive convenience that keyless cars originally promised.
   The Mitigation: This friction is an intentional security design choice. It defeats real-world crimes where a thief steals both the keys and the phone simultaneously (gym locker, luggage, hotel room). The conscious action takes

place while the driver is settling into the seat and fastening their seatbelt, reducing perceived delay to negligible levels.

3. **Limitation: Secondary User Device Battery Depletion**
   The Challenge: If a driver's smartphone battery is dead or the device is lost or damaged, the primary acoustic channel is unavailable.
   The Mitigation: The system implements the State-Gated Dormant Fallback Mode. Once the 30-to-45 second Stage 1 acoustic window times out, the manual fallback PIN interface awakens, ensuring an authentic driver with a flat battery is never stranded.

4. **Limitation: Open Window & Structural Boundary Compromise**
   The Challenge: Acoustic boundary containment relies on the structural transmission loss of laminated safety glass and elastomeric weatherstripping. If a driver leaves vehicle windows rolled down in an unmonitored parking space, acoustic energy can escape into the immediate perimeter.
   The Mitigation:
   - Near-ultrasonic frequencies (16.0 kHz to 17.0 kHz) experience rapid inverse-square divergence and high physical air attenuation.
   - Transmit amplitudes are scaled to unit levels, restricting the decipherable signal-to-noise ratio (SNR) perimeter strictly to the immediate interior/near field.
   - The ECU enforces a strict 30-to-45 second ingress timer; the microphone is not powered indefinitely.

5. **Limitation: Acoustic Shadowing, In-Cabin Multi-Path & Environmental Distortion**
   The Challenge: Heavy winter clothing covering a driver's pocket, seat foam absorption, or unusual passenger placement can create acoustic standing wave nulls. Furthermore, low-end smartphone speakers might struggle with non-linear distortion at high frequencies.
   The Mitigation:
   - Carrier frequencies are fixed at Pair A (16.0 kHz and 17.0 kHz), well within the Nyquist rate of commodity 44.1 kHz and 48.0 kHz audio DACs.
   - The modulation uses Continuous-Phase Frequency-Shift Keying (CP-FSK) with a 10.0 ms bit period (100 baud), ensuring long symbol integration times.
   - The ECU applies non-coherent Quadrature (I/Q) energy demodulation, making recovery immune to phase shifts.
   - In vehicles equipped with multiple microphones (telematics mic, ANC headliner mics), the ECU performs spatial diversity combining (SNR Selection or Maximal Ratio Combining).

IMPORTANT ENGINEERING NOTICE: DEMO VS. PRODUCTION TIMING

In this interactive demonstration build, the [Manual Override] button is made immediately accessible on the primary interface. This design choice is strictly intended to facilitate rapid testing and allow evaluators to inspect the manual fallback and hidden technician gateways without enforcing real-time delays.

In a production automotive deployment (per Patent Claims 8, 12, 24, and 25):
- **State-Gated Dormancy:** The manual fallback interface remains completely unpowered, unclocked, and software-isolated during Stage 1.
- **Mandatory Temporal Delay:** The driver cannot access manual PIN entry or the technician unmasking gateway until the 30-to-45 second primary acoustic listening window experiences an explicit timeout.
- **Parallel Brute-Force Elimination:** This ensures an attacker inside the cabin cannot execute parallel PIN attacks while the ECU is evaluating acoustics.

SECTION 2: THREAT MODELING & COMPETITIVE DIFFERENTIATION

2.1 Threat Modeling & Real-World Attack Vectors

Vehicle theft has shifted decisively away from mechanical forced entry toward electronic exploitation of radio frequency (RF) vulnerabilities and opportunist physical theft. In alignment with ISO/SAE 21434 threat analysis and real-world criminology data, QAL is specifically architected to defeat five primary threat vectors:

PRIMARY VEHICLE THREAT TAXONOMY

THREAT 1: RF Relay Station Attack (RSA)	–> Extends RF signal from house to driveway
THREAT 2: Letterbox & Hallway Key Theft	–> Physical theft of key fob only
THREAT 3: "Gym Locker" Multi-Asset Theft	–> Theft of BOTH key fob and phone together
THREAT 4: Remote Cellular Audio Streaming	–> Digital audio bridge streamed into cabin
THREAT 5: OBD-II Key Cloning / CAN Inject	–> Diagnostic port / bus spoofing attack

Vector 1: Wireless Relay Station Attacks (RSA)

- **The Threat Mechanism:** In a classic two-person relay attack, Actor A positions an RF amplifier close to the exterior wall or front door of an owner's residence to detect the passive low-frequency (LF) polling signal of a key fob kept inside. Actor A establishes an ultra-high-frequency (UHF) wireless bridge to Actor B standing next to the vehicle on the driveway. The vehicle's Body Control Module (BCM) falsely concludes that the authentic key fob is adjacent to the door handle, unlocking the doors and authorizing ignition.
- **Why Conventional Systems Fail:** Radio frequency signals (315 MHz, 433 MHz, 2.4 GHz) pass directly through residential brick walls, double-glazed home windows, vehicle glass, and steel body panels.
- **How QAL Mitigates the Threat:** The primary RF bridge only succeeds in waking the ECU into Stage 1. Propulsion interlocks (starter motor, inverter gate drivers, high-voltage contactors) remain immobilized. Because the secondary authentication token is near-ultrasonic acoustic energy (16.0 kHz to 17.0 kHz), an external RF amplifier cannot relay it. Furthermore, the acoustic boundary of the vehicle (laminated safety glass and rubber weatherstripping) prevents the token from being captured outside the vehicle.

Vector 2: Letterbox Fishing & Residential Key Burglary

- **The Threat Mechanism:** According to police intelligence, opportunistic thieves frequently fish car keys off hallway tables through letterboxes using telescopic rods, or conduct quick residential break-ins purely to locate vehicle keys.
- **Why Conventional Systems Fail:** In standard single-factor vehicles, physical possession of the authentic key fob grants 100% unrestricted vehicle operation. Once the key is in hand, the vehicle cannot distinguish between the legitimate owner and an intruder.
- **How QAL Mitigates the Threat:** Possession of the physical key fob alone only unlocks the vehicle doors and wakes the ECU into Stage 1. Once seated inside, the vehicle remains fully immobilized. Without the secondary authorized user device and the owner's private knowledge credential, the vehicle cannot be driven away, rendering the stolen key useless on its own.

Vector 3: Simultaneous Key and Smartphone Theft ("The Gym Locker" Vector)

- **The Threat Mechanism:** In sports facilities, health clubs, luggage transit, hotel rooms, or public cloakrooms, an opportunistic thief often steals a jacket or bag containing both the physical car key and the driver's smartphone together.
- **Why Modern Digital Keys Fail:** Mass-market digital key architectures (including passive BLE and background phone keys) operate hands-free. If a thief places the victim's stolen bag on the passenger seat, the car detects the phone in proximity and authorizes drive-away immediately.
- **How QAL Mitigates the Threat:** QAL deliberately rejects passive background transmission. Under QAL's Proof-of-Intent model, acoustic token emission requires an intentional, conscious user action: the driver must unlock the mobile device via biometric security (FaceID, fingerprint, or lockscreen passcode) and manually trigger the transmission. Even if an unauthorized party holds both the authentic key fob and the authentic phone, they cannot trigger the acoustic transmission without knowing the device unlock credential.

Vector 4: Remote Cellular Audio Streaming & Network Bridging

- **The Threat Mechanism:** An adversary might attempt to defeat in-cabin acoustic authentication by leaving an accomplice's mobile device or small speaker hidden inside the vehicle cabin while an accomplice streams recorded audio over a cellular VoIP call or cellular data connection.
- **Why the Attack Fails Against QAL:**
  - **Codec Low-Pass Filtering:** Standard cellular voice codecs (AMR, AMR-WB, G.711) and mobile VoIP bridges enforce aggressive low-pass acoustic filtering designed for human speech, heavily cutting off frequencies above 3.4 kHz (narrowband) or 7.0 kHz (wideband). Near-ultrasonic frequencies between 16.0 kHz and 17.0 kHz are completely stripped by cellular network encoders.
  - **Latency Bounding:** In bidirectional challenge-response configurations (Patent Claim 18), the vehicle ECU hardware timer measures the acoustic round-trip latency from the end of the challenge chirp to the arrival of the response. Cellular and Internet data hops introduce network transport delays typically ranging from 250 ms to over 1000 ms, exceeding the strict round-trip threshold ($\Delta t \leq 1.5$ s).

Vector 5: OBD-II Port Key Programming & CAN Bus Injection

- **The Threat Mechanism:** When thieves cannot execute a relay attack, they frequently smash a small quarterlight window or pick the door lock to gain cabin access. Once inside, they plug an unauthorized handheld key programming tool into the standard on-board diagnostics (OBD-II) port under the steering column, coding a blank key fob in under 60 seconds. Alternatively, organized gangs peel back the front bumper to tap into exposed headlight CAN bus wiring, injecting spoofed "key-present" CAN frames to bypass the factory alarm.
- **Why Conventional Systems Fail:** Factory immobilizers trust the vehicle's diagnostic bus. Once a rogue tool commands the Body Control Module (BCM) to register a new key, or CAN messages are successfully injected, the car treats the thief as an authorized owner and allows immediate drive-away.
- **How QAL Mitigates the Threat:** The QAL immobilizer architecture operates as an independent security firewall separated from standard OBD key-registration tables. Even if a thief successfully codes a blank RF key fob via the diagnostic port or injects spoofed "key-valid" CAN messages to wake the system, the vehicle only advances to Stage 1. Vehicle propulsion contactors, high-voltage relays, and starter circuits remain hard-locked because the ECU requires the independent, cryptographically signed acoustic token.

2.2 Competitive Differentiation Matrix

Architectural Metric	Conventional PKES (Key Fob)	Ultra-Wideband (UWB ToF)	Touchscreen PIN-to-Drive	Ambient Sound (Choi et al.)	QAL Acoustic Architecture
RF Relay Attack Immunity	Vulnerable (Primary target)	High (ToF physical distance checks)	High (Requires knowledge factor)	Moderate (Noise sensitive)	**High** (Physical cabin boundary attenuation)
Stolen Physical Key Protection	None (Drive-away granted)	None (Stolen key calculates legitimate distance)	High (Requires independent PIN)	None (Key is physically inside cabin)	**High** (Requires secondary in-cabin token)
Simultaneous Phone + Key Theft	Vulnerable	Vulnerable (Passive link authorizes start)	High (Requires separate PIN)	Vulnerable (Passive correlation authorizes start)	**High** (Biometric device unlock required before emit)
BOM / Hardware Cost Impact	Baseline standard	High (GBP 150-300 in chips, antennas, cabling)	Low (Software on pre-existing screen)	Low (Uses pre-existing cabin microphones)	**Low** ($0 added vehicle hardware; re-uses telematics mic)
Display Smudge Attack Resistance	N/A	N/A	Vulnerable (Oily fingerprint traces on glass)	N/A	**Extremely High** (Zero routine vehicle screen touch)
Offline Autonomy (No 4G/Cloud)	100% Offline	100% Offline	100% Offline	Variable (Fails in noisy environments)	**100% Local Execution** (Local ECU & phone processing)
Driver UX Friction Level	Zero friction (Completely passive)	Zero friction (Completely passive)	High friction (Mandatory manual typing every trip)	Low (Unpredictable correlation failure rate)	**Intentional / Low** (2-3s tap while fastening seatbelt)

2.3 Technical Differentiation from Specific Prior Art Families

To provide technical clarity for patent examiners and automotive IP counsel, QAL is explicitly distinguished from previously cited automotive and acoustic patent literature:

1. **Differentiation from Automotive Ultrasonic Time-of-Flight (Volvo Family: US 10,759,383 B2 / US 11,528,612 B2 / US 11,974,133 B2):**
   - **The Prior Art:** The Volvo patent family describes systems utilizing exterior ultrasonic transducers (such as exterior ultrasonic parking bumper sensors) to calculate physical distance ranging and echo reflections between an approaching driver and the exterior vehicle perimeter.
   - **The QAL Distinction:** QAL operates completely independently of acoustic distance calculation, Time-of-Flight ranging, echo reflections, or exterior transducers. QAL requires zero exterior acoustic hardware. Instead, QAL relies entirely on the structural transmission loss and spatial boundary containment of an enclosed cabin (glass, sheet metal, and weatherstripping) to physically trap near-ultrasonic data payloads emitted by a handheld device.
2. **Differentiation from Ambient Acoustic Correlation Schemes (Choi et al., 2018):**
   - **The Prior Art:** Choi, Seo, and Lee (Journal of Advanced Transportation, 2018) propose sampling ambient environmental background noise simultaneously on a key fob and a vehicle to establish acoustic co-presence.
   - **The QAL Distinction:** Ambient background sampling is notoriously unreliable in real-world conditions; rain, wind, passing heavy goods vehicles, or quiet residential garages cause severe correlation mismatches. Furthermore, Choi et al. rely on audible chirps that disturb occupants. In contrast, QAL transmits an inaudible, deterministic Continuous-Phase Frequency-Shift Keying (CP-FSK) cryptographic bitstream (16.0 kHz to 17.0 kHz) steganographically masked beneath a low-pass-filtered vehicle chime, backed by an unpowered dormant fallback state machine and an encrypted ECU vault.
3. **Differentiation from Commercial Ultrasonic Point-of-Sale Systems (Malbec Labs: US 9,098,865 B2):**
   - **The Prior Art:** US Patent 9,098,865 describes near-field communication utilizing unmasked ultrasonic bursts (16.0 kHz to 22.0 kHz) for commercial point-of-sale (POS) payment transactions in open retail spaces.
   - **The QAL Distinction:** Retail POS systems broadcast unmasked, raw ultrasonic pulses across open retail air. QAL is an automotive powertrain immobilization firewall engineered to operate inside a closed acoustic resonant cavity (the passenger cabin). QAL implements client-side steganographic masking beneath an audible carrier, enforces a steep 12th-order Butterworth low-pass filter with a 1.5 kHz silent guard band, and directly gates ECU starter relays, high-voltage battery contactors, and Shift-by-Wire mechanical locks.
4. **Differentiation from In-Cabin Ultrasonic Positioning Prior Art (US Patent Application 2019/0025402 A1):**
   - **The Prior Art:** US 2019/0025402 A1 describes systems utilizing ultrasonic acoustic signals (16 kHz to 26 kHz) to calculate the spatial coordinates or seating zone of a smartphone inside a car (e.g. distinguishing whether a device is in the driver seat vs. passenger seat) via acoustic time-difference ranging.
   - **The QAL Distinction:** QAL operates completely independently of device localization, spatial triangulation, or acoustic distance measurement. QAL does not care which seat or pocket the phone is in. It utilizes in-cabin acoustic transmission strictly as an out-of-band cryptographic authentication transport layer to release multi-stage ECU immobilizer interlocks.
5. **Differentiation from General Audible Audio Token Prior Art (Qualcomm: US 7,533,735 B2 & US 7,966,497 B2):**
   - **The Prior Art:** US Patent 7,966,497 B2 (Gantman & Rose, assigned to Qualcomm) describes a hand-held hardware token emitting an acoustic or wireless signal representing a digital signature from a private/public key pair, combined with a manual PIN entered into a fixed terminal (such as a bank ATM or financial POS) for account verification.
   - **The QAL Distinction:** Qualcomm's architecture is tailored for banking terminals, ATMs, and financial retail transactions using asymmetric public/private key handshakes combined with manual PINs. QAL is an automotive powertrain immobilization firewall engineered to operate inside a closed acoustic resonant cavity (the vehicle passenger cabin). QAL utilizes symmetric PBKDF2/AES-256 key vaults, hardware security module (HSM) root secrets, and time-decayed ephemeral tokens to gate vehicle starter relays, high-voltage battery contactors, and Shift-by-Wire mechanical locks, operating independently of financial networks or banking backends.

2.4 Acoustic Jamming & Denial-of-Service (DoS) Threat Model

The Threat Scenario: An adversary positions an external acoustic transducer near the target vehicle, broadcasting continuous high-amplitude broadband noise or targeted acoustic tones in the 16.0 kHz to 17.0 kHz band to corrupt the in-cabin microphone signal.

System Resilience & Fallback Behavior:

1. **Acoustic Boundary Attenuation:** Laminated glass and EPDM weatherstripping attenuate exterior noise by 30 dB to 42 dB, requiring an attacker to emit impractically loud exterior sound (> 105 dB SPL) to disrupt in-cabin near-field transmission.
2. **Graceful Degradation to Dormant 2FA Fallback:** If in-band jamming corrupts the acoustic bitstream (triggering DTC 0x04 CRC mismatch) and the Stage 1 timer expires (DTC 0x01), the ECU awakens the dormant manual fallback interface.
3. **Security Invariant Preserved:** Even under total acoustic jamming, the vehicle remains fully immobilized until the driver enters the correct manual fallback PIN alongside physical key possession. Jamming can cause temporary convenience denial, but can never force unauthorized vehicle unlock or drive-away.

2.5 ISO/SAE 21434 TARA (Threat Assessment and Remediation Analysis)

In compliance with ISO/SAE 21434 cybersecurity engineering requirements, the threat vectors addressed by QAL are scored under the formal Threat Assessment and Remediation Analysis (TARA) methodology:

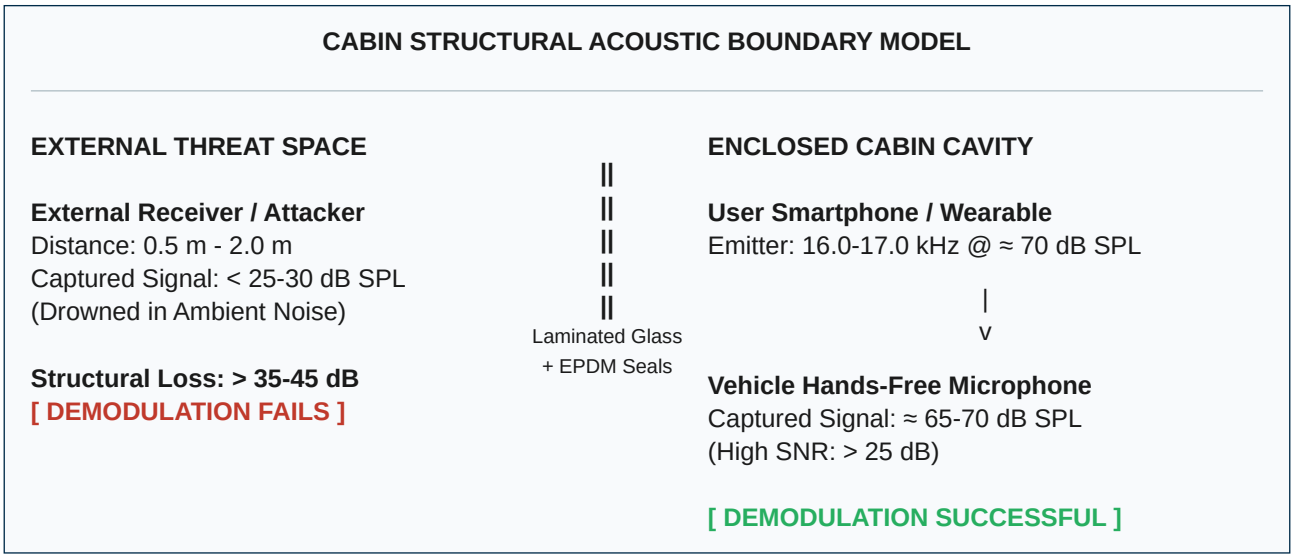
Threat Scenario	Asset / Target	STRIDE Type	Attack Feasibility	Impact Rating	Residual Risk	Primary QAL Security Control
RF Relay Attack (RSA)	Starter / HV Contactors	Elevation of Privilege	High (Standard Tools)	Severe (Total Loss)	**Low** (Negligible)	Physical acoustic boundary containment + Out-of-band Factor 2 gating.
Letterbox / Stolen Key Fob	Powertrain Interlock	Spoofing	Medium (Physical Entry)	Severe (Total Loss)	**Low** (Negligible)	Independent secondary acoustic token gating + Owner secret isolation.
Simultaneous Phone + Key Theft	Mobile App + Fob	Elevation of Privilege	Medium (Locker Break-in)	Severe (Total Loss)	**Low** (Negligible)	Proof-of-Intent: Biometric lockscreen (FaceID/PIN) required to emit token.
OBD-II Key Programming	BCM Key Register	Tampering	High (Handheld Hackers)	Severe (Total Loss)	**Low** (Negligible)	Independent ECU immobilizer firewall; Stage 2 locked to HSM vault.
Cellular Audio Stream Relay	Interior Microphone	Spoofing / Relay	Low (Specialized Setup)	Moderate (Unauthorized Start)	**Low** (Negligible)	Codec filtering (> 14 kHz cut) + Round-trip latency limit ($\Delta t \leq 1.5$ s).
In-Band Acoustic Jamming	Ingress Listening	Denial of Service	Medium (Near-Field Noise)	Minor (Friction / Delay)	**Low** (Acceptable)	State-Gated Dormant Fallback: Reverts to 2FA Physical Key + Manual PIN.

SECTION 3: PHYSICAL & ACOUSTIC LAYER SPECIFICATION & BOUNDARY TESTING PROTOCOL

3.1 Acoustic Boundary Containment Physics

The core physical security of the QAL system relies on structural transmission loss, atmospheric attenuation, and spatial confinement within an enclosed automotive cabin boundary (Patent Claim 1 and Claim 20).

Unlike radio frequency (RF) signals, which penetrate non-conductive barriers with negligible attenuation, near-ultrasonic acoustic waves (16.0 kHz to 17.0 kHz) experience severe physical-layer decay when encountering modern automotive enclosures:



The Four Physical Mechanisms of Confinement:

1. **Laminated Safety Glass Acoustic Impedance Mismatch:**
   Automotive windscreens and modern acoustic side-laminated windows consist of two glass sheets bonded by a viscoelastic Polyvinyl Butyral (PVB) acoustic interlayer. While low-frequency rumble passes through glass, high frequencies around 16.0 kHz suffer significant reflection at the air-glass interface and severe shear-wave damping inside the core polymer. Transmission loss through acoustic laminated glass at 16.0 kHz typically exceeds 35 dB to 42 dB.
2. **Elastomeric EPDM Door Weatherstripping:**
   Automotive doors utilize dual-bulb or triple-lip Ethylene Propylene Diene Monomer (EPDM) hollow rubber seals compressed between the door frame and the steel chassis. High-frequency sound waves cannot easily diffract around these compressed elastomeric barriers, eliminating acoustic air leakage paths.
3. **Atmospheric Molecular Absorption (Air Attenuation):**
   Acoustic energy absorption in ambient air increases quadratically with frequency. At 20 degrees Celsius and 50% relative humidity, air absorption at 16.5 kHz is approximately 0.5 dB per meter. Combined with standard inverse-square spherical spreading (6 dB loss per doubling of distance), signal amplitude drops precipitously once outside the glass.
4. **Ambient Ultrasonic Noise Floor Masking:**
   In real-world outdoor environments (driveways, parking lots, roadsides), ambient environmental noise (tire friction, wind turbulence, leaf rustling, distant traffic, and insect activity) generates a continuous background acoustic noise floor of 35 dB to 50 dB SPL across high frequencies. Once an acoustic token leaks through the glass and is attenuated by target structural transmission loss margins (30 dB to 42 dB), its received power falls well below the ambient noise floor (SNR < 0 dB), preventing reliable bitstream demodulation.

(Target Empirical Benchmarks: Expected attenuation of 30 dB to 42 dB through automotive laminated glass and weatherstripping, subject to physical test rig validation).

3.2 Digital Signal Processing (DSP) & Modulation Pipeline

1. Continuous-Phase Frequency-Shift Keying (CP-FSK)

To eliminate abrupt phase discontinuities between adjacent bits that cause wideband acoustic splatter and audible transient clicks, QAL utilizes Continuous-Phase Binary Frequency-Shift Keying (Patent Section 6.2 and Claim 1).

The instantaneous phase $\theta(t)$ is continuously accumulated across symbol transitions:

$$\theta(t) = 2\pi \int_0^t f(\tau)\,\mathrm{d}\tau$$

In Plain English: Drawing an Unbroken Wave (Continuous-Phase Audio)

- **The Problem ("Torn" Audio Waves):** In ordinary digital audio, switching from 16.0 kHz to 17.0 kHz abruptly cuts the waveform. Like lifting a pen off paper mid-stroke, this phase discontinuity produces an audible click and splatters unwanted energy into neighboring frequencies.
- **What the Formula Does (Never Lifting the Pen):** The integral tracks the wave's exact instantaneous position. When transitioning bits, the waveform continues smoothly from its current phase, simply adjusting its rate of oscillation without jumping.
- **Why It Matters:** The acoustic energy remains strictly confined within the designated carrier bins without bleeding into adjacent audible bands.

Carrier Frequencies (Primary Pair A):

- Binary '0' (f_0): 16,000 Hz
- Binary '1' (f_1): 17,000 Hz
- Carrier Separation (Δf): 1,000 Hz (Orthogonal signaling)

Alternative Pairs Defined in Patent Claim 4:

- Pair B: 18,000 Hz / 19,000 Hz
- Pair C: 18,500 Hz / 19,500 Hz
- Pair D: 20,000 Hz / 21,500 Hz (Ultrasonic band for 48/96 kHz hardware)

Symbol Timing & Hardware Sampling Rate (Patent Claim 5 - Updated Specification):

- Bit Duration (T_b): 10.0 ms (100 baud symbol rate).
- Nominal Reference Sampling Rate ($f_{s,\text{ref}}$): 44,100 Hz (CD audio reference standard).
  - Nominal Samples per Bit ($N_{s,\text{ref}}$): $44,100 \times 0.010 = 441$ samples.
- Automotive Telematics Operational Rate ($f_{s,\text{auto}}$): 48,000 Hz (Automotive ADC default).
  - Operational Samples per Bit ($N_{s,\text{auto}}$): $48,000 \times 0.010 = 480$ samples.

Engineering Implementation Alignment:

While baseline mathematical proofs and legacy media files are defined at the 44.1 kHz reference (441 samples/window), automotive production audio hardware, infotainment operating systems (Android Automotive / QNX), and the live interactive demonstration suite execute at the standard automotive telematics rate of **48.0 kHz**. Under 48 kHz sampling, the non-coherent Quadrature (I/Q) energy detector integrates across an exact **480-sample symbol window**, preserving the continuous-phase 100-baud orthogonal modulation profile without phase drift or spectral widening.

2. Envelope Shaping (Single Burst Fade vs. Per-Bit Windowing)

Applying a Hann window to every individual 10 ms bit pulse introduces a 100 Hz amplitude-modulation sideband that leaks energy into the silent guard band.

In production (Patent Section 6.2), the acoustic burst maintains a constant unit amplitude envelope across all internal symbol transitions. A raised-cosine taper (5.0 ms fade-in and fade-out) is applied once across the absolute beginning and end of the complete transmission to prevent digital speaker clicks:

$$w[n] = 0.5 - 0.5\cos\left(\frac{\pi n}{N_{\text{fade}}}\right), \quad 0 \le n < N_{\text{fade}}$$

In Plain English: The 5-Millisecond Volume Dimmer (Eliminating Speaker Clicks)

- **The Problem (The Digital "Pop"):** Instantly starting audio from pure digital silence forces the speaker cone to snap forward violently, creating a wideband click that splashes noise into the guard band.

- **What the Formula Does:** Acts as a high-speed volume dimmer over exactly 5.0 milliseconds (0.005 s) at the very start and end of the transmission, allowing the speaker cone to glide smoothly into motion.

3. 12th-Order Butterworth Guard Band Low-Pass Filter

When the near-ultrasonic payload is mixed beneath an audible welcome chime (such as a pleasant 1.0 kHz to 3.0 kHz chord), the chime is pre-processed through a digital low-pass filter to prevent audible harmonics from bleeding into the receiver detection bins (Patent Claim 3):

- **Filter Topology:** 12th-order Butterworth Low-Pass Filter implemented as a cascade of six second-order section (SOS) biquads.
- **Cutoff Frequency (f_c):** 14,000 Hz
- **Stopband Attenuation:** > 80 dB attenuation at 16,000 Hz
- **Unmodulated Spectral Guard Band:** ≥ 1.5 kHz wide (silent zone between 14.0 kHz and 16.0 kHz)

STEGANOGRAPHIC SPECTRAL PROFILE

Audible Carrier Chime	Silent Guard Band	Ultrasonic CP-FSK Data
Frequency: 0 Hz to 14.0 kHz	Frequency: 14.0 to 16.0 kHz	Frequency: 16.0 to 17.0 kHz
Peak Amplitude: 0.65	Rolloff: > 80 dB (12th-Order)	Peak Amplitude: 0.25
Status: Filtered Chime Audio	Status: Unmodulated Silence	Status: Encrypted 2FA Payload

4. Composite Signal Scaling (Patent Claim 15)

To prevent clipping and digital-to-analog converter (DAC) distortion, composite signal amplitudes are explicitly constrained:

- **Audible Chime Peak Gain:** ≈ 0.65
- **Near-Ultrasonic Data Peak Gain:** ≈ 0.25
- **Maximum Composite Waveform Peak:** ≤ 0.90 of full-scale ($0.65 + 0.25 = 0.90$)

5. Receiver Demodulation via Non-Coherent Quadrature (I/Q) Energy Detection

To guarantee robust demodulation regardless of phone orientation, multi-path reflections, or arbitrary acoustic phase shifts, the ECU receiver executes non-coherent I/Q detection (Patent Section 6.4). For each symbol window (N_s samples):

$$I_0 = \sum_{n=0}^{N_s-1} x[n] \sin\left(\frac{2\pi f_0 n}{f_s}\right), \quad Q_0 = \sum_{n=0}^{N_s-1} x[n] \cos\left(\frac{2\pi f_0 n}{f_s}\right)$$

$$I_1 = \sum_{n=0}^{N_s-1} x[n] \sin\left(\frac{2\pi f_1 n}{f_s}\right), \quad Q_1 = \sum_{n=0}^{N_s-1} x[n] \cos\left(\frac{2\pi f_1 n}{f_s}\right)$$

Symbol Energy (Magnitude-Squared):

$$M_0 = I_0^2 + Q_0^2, \quad M_1 = I_1^2 + Q_1^2$$

Note that M is the magnitude-squared (energy) quantity rather than the square-rooted magnitude. Because the square root is a monotonic transform, the comparison $M_1 > M_0$ is identical under either formulation, and omitting the root avoids an unnecessary per-symbol operation on the embedded target.

Bit Decision Rule:

$$\text{Decoded Bit} = \begin{cases} 1 \text{ if } M_1 > M_0 \\ 0 \text{ if } M_0 \geq M_1 \end{cases}$$

In Plain English: How the Car Decodes the Sound

- **Step 1 (The "I and Q" Math):** Sound bounces off windows and seats, potentially reaching the microphone inverted or phase-shifted. The ECU evaluates each frequency from two orthogonal perspectives simultaneously (In-Phase I and Quadrature Q).

- **Step 2 (Calculating True Energy):** Combining the orthogonal measurements ($I^2 + Q^2$) reveals the true physical energy of the tone regardless of its arrival phase.
- **Step 3 (The Decision Rule):** The ECU compares energy levels: if $M_1 > M_0$, it records a binary 1; otherwise, a binary 0. Executing this evaluation every 10 milliseconds reconstructs the cryptographic security token with 0% Bit Error Rate.

3.3 Physical Testing Protocol (Acoustic Boundary Attenuation)

To physically validate near-ultrasonic boundary containment and provide empirical proof for patent examiners and automotive engineers, the following testing procedure should be executed:

1. Required Test Equipment

- **Emitter:** Standard consumer smartphone or portable speaker capable of 44.1 kHz / 48.0 kHz playback.
- **Acoustic Receiver / Analyzer:** Calibrated measurement microphone with flat high-frequency response up to 20.0 kHz (e.g., miniDSP UMIK-1, Dayton Audio EMM-6, or an automotive telematics mic connected to an audio analyzer).
- **Spectrum Analysis Software:** Audacity, REW (Room EQ Wizard), or a real-time FFT spectrum analyzer.
- **Test Barrier:** A production vehicle equipped with factory laminated windscreen and closed rubber door seals (or a bench test rig holding automotive laminated glass and compressed EPDM rubber gaskets).

2. Step-by-Step Test Procedure

1. **Calibration:** Place the measurement microphone 0.5 meters from the emitting device in free air. Play the generated token file. Adjust the volume until the 16.0 kHz and 17.0 kHz carrier peaks read between 65 dB and 72 dB SPL.
2. **In-Cabin Baseline Measurement:** Place the emitter on the driver's seat and position the receiving microphone in the roof console (mimicking the factory telematics hands-free mic). Close all doors and windows. Record the transmission and verify:
   - Carrier SNR ≥ 20 dB above cabin background noise.
   - Frame demodulation passes with zero Bit Error Rate (BER = 0%).
3. **Exterior Boundary Attenuation Measurement:** Keep the emitter playing at the identical volume inside the vehicle. Move the measurement microphone outside the vehicle:
   - **Test Point A:** 5 cm outside the driver's side laminated window.
   - **Test Point B:** 5 cm outside the rubber door frame weatherstripping.
   - **Test Point C:** 1.0 meter away from the vehicle door (representing a thief with an RF relay box).
4. **Data Recording:** For each exterior test point, capture the FFT spectrum and record:
   - Received peak SPL at 16.0 kHz and 17.0 kHz.
   - Ambient exterior noise floor level.
   - Structural Transmission Loss: $\mathrm{TL} = \mathrm{SPL}_{\mathrm{inside}} - \mathrm{SPL}_{\mathrm{outside}}$.
   - Demodulation Attempt: Run recorded exterior audio through the decoder software.

3. Expected Engineering Results & Target Performance Criteria

- **In-Cabin Capture:** The system registers valid Start-of-Frame (SOF), verified CRC32 checksum, and Stage 2 Authorization under normal in-cabin operating conditions.
- **Exterior Boundary Loss (Target Engineering Benchmark):** Expected structural transmission loss (TL) across closed automotive laminated glass and elastomeric weatherstripping is targeted between 30 dB and 42 dB.
- **Exterior Demodulation (Pass Criterion):** Demodulation of exterior recordings fails completely (CRC32 validation fails, SOF undetected, or carrier peaks buried below the ambient noise floor). This demonstrates that structural transmission loss suppresses signal levels below the demodulation threshold of external receivers.

SECTION 4: ECU CRYPTOGRAPHIC ARCHITECTURE & STATE MACHINE

4.1 Multi-Stage Operational State Machine & Powertrain Interlocks

In strict alignment with Patent Claims 1, 8, 9, and 20, the vehicle Electronic Control Unit (ECU) manages vehicle propulsion through a deterministic, multi-stage state machine:

ECU MULTI-STAGE OPERATIONAL STATE MACHINE

[STATE 0: LOCKED] (400)
- Powertrain Interlocks: DISABLED (High-voltage contactors & starter relays isolated)
- In-Cabin Microphones: UNPOWERED (0V)
- Manual Fallback UI: DORMANT & UNPOWERED

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v Preliminary Access Event (Door handle touch / RF fob ingress, 402)

[STAGE 1: PHYSICAL-PRESENCE LISTENING] (410)
- Cabin Microphones: ENERGIZED (Listening window: 30 to 45 seconds at 48.0 kHz)
- Powertrain Interlocks: STRICTLY INHIBITED
- User Retry Loop (424): Pressing Start button refreshes timer without penalty

Path A: Valid Token Captured (422)
–> **[STAGE 2: AUTHORIZED] (430)**
- High-voltage contactors closed
- Starter relay circuit enabled
- Propulsion released for driving

Path B: Window Times Out (426)
–> **[DORMANT FALLBACK] (440)**
- In-cabin manual PIN screen awakens
- 3 Failed entries –> LOCKOUT (450)
- 3 Empty entries –> SERVICE (470)

The Four Discrete Operating States:

1. **State 0: Locked State (400)**
   The baseline quiescent state when the vehicle is parked. High-voltage (HV) main traction contactors, fuel pump relays, electronic transmission park-locks (Shift-by-Wire), and starter motor solenoids are physically isolated or software-inhibited. The in-cabin microphones and manual fallback interfaces remain completely unpowered.

2. **Stage 1: Physical-Presence Listening State (410)**
   Triggered by a preliminary access or wake-up event (such as opening the driver door, sitting in the driver's seat, or detecting a primary wireless key fob). The ECU energizes the in-cabin telematics microphone and initiates an ingress timer (configurable between 30.0 and 45.0 seconds). Crucially, powertrain propulsion interlocks remain completely disabled throughout Stage 1.

3. **Stage 2: Starter / Propulsion Authorization State (430)**
   Transitioned to if, and only if, the in-cabin microphone captures a valid, cryptographically verified acoustic token within the Stage 1 window. Upon successful HMAC-SHA256 signature verification and freshness confirmation, the ECU commands the closure of high-voltage contactors and enables starter circuits, permitting vehicle operation.

4. **Maintenance / Service Mode (470)**
   An extended diagnostic session conforming to ISO 14229 UDS Service 0x27 (SecurityAccess). Enables workshop technicians to temporarily bypass acoustic immobilizer checks for servicing without possessing the owner's private credentials.

4.1.1 Functional Safety & In-Motion Protection (ISO 26262 ASIL-B/D Design Target)

A primary safety requirement under ISO 26262 is the prevention of unintended powertrain deactivation while driving:

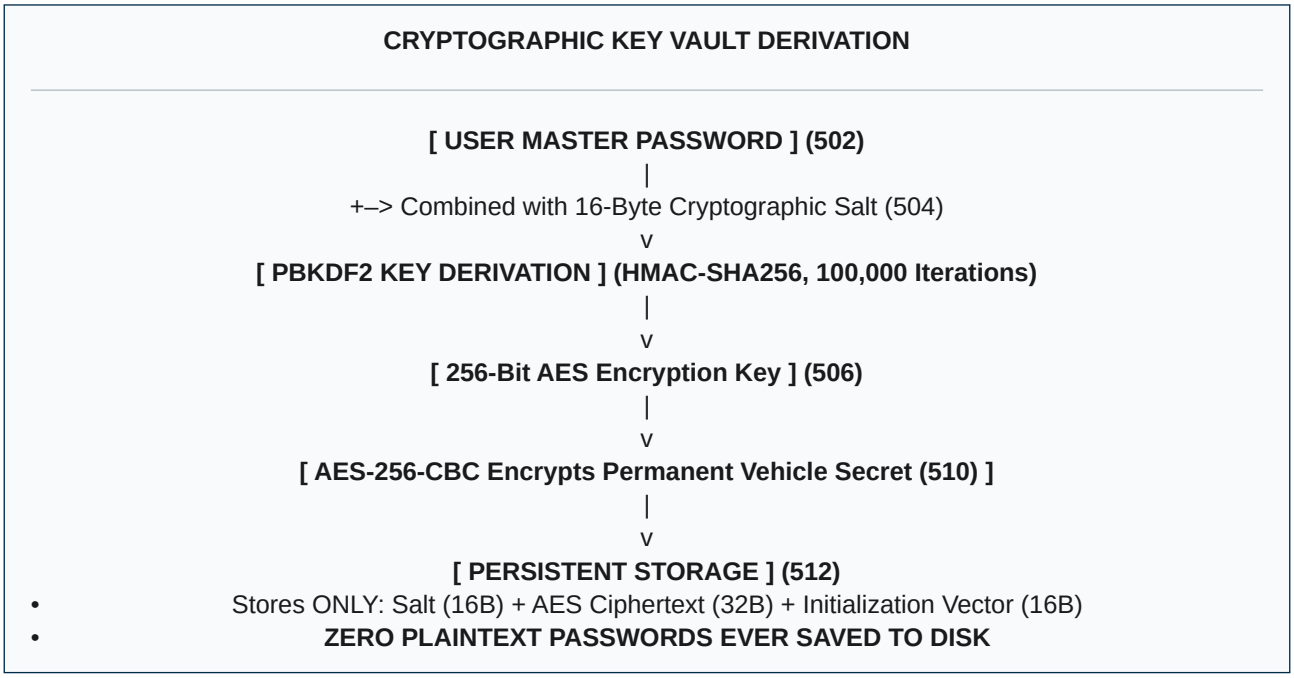
- **Standstill Gating Only (Fail-Secure at Rest):** The QAL security state machine operates strictly prior to vehicle motion. Gating high-voltage contactors, fuel pump relays, and starter solenoids is executed solely to authorize initial engine start or high-voltage pre-charge.
- **No In-Motion Interruption (Fail-Operational / Fail-Silent in Motion):** Once the vehicle transitions to Stage 2 Authorized and the powertrain is engaged, the acoustic subsystem is powered down. The ECU contains a

hardcoded safety interlock ensuring that no acoustic timeout, microphone fault, or secondary token failure can open contactors or cut vehicle power while vehicle velocity exceeds 0 km/h.

User Start-Button Retry Loop (Patent Claim 9 / [0091]):
If a driver takes longer than 45 seconds to fasten their seatbelt, or if acoustic playback is delayed, the driver does not suffer an anti-theft penalty. Pressing the vehicle's engine Start button executes an immediate User Retry Loop (424):
- Refreshes the Stage 1 timer for an additional 30 to 45 seconds.
- Re-energizes the in-cabin microphone.
- Clears any transient capture errors without locking out the legitimate driver.

4.2 Cryptographic Key Vault (Zero Plaintext Persistence)

In strict compliance with Patent Claim 17, Patent Section 6.8, and modern automotive cybersecurity standards (EVITA / ISO 21434), the vehicle ECU and paired mobile applications are architected so that the plaintext master password is never written to non-volatile disk storage:



The Derivation Pipeline (Step-by-Step):

1. **Password Salt Generation:** During initial onboarding, a cryptographically secure pseudo-random number generator creates a 16-byte random salt.
2. **PBKDF2 Derivation:** The user's chosen password and the 16-byte salt are processed using Password-Based Key Derivation Function 2 (PBKDF2) configured with HMAC-SHA256 and 100,000 iterations to derive a high-entropy 256-bit AES encryption key.
3. **Vehicle Secret Encapsulation:** This derived 256-bit AES key is used to encrypt the car's permanent 256-bit root Vehicle Secret via AES-256-CBC with a randomized Initialization Vector (IV).
4. **Persistent Storage:** Only the cryptographic salt, the IV, and the encrypted ciphertext are persisted in local flash storage. Extracting raw memory from an ECU or smartphone yields zero actionable plaintext keys.
5. **Vehicle Identification Binding (VIN Binding):** To prevent an authentic acoustic token recorded from one vehicle from ever functioning in another vehicle of the same model, the Vehicle Secret is bound to the Vehicle Identification Number (VIN) within the Hardware Security Module (HSM):

$$\text{Pairing Certificate} = \text{HMAC-SHA256}(\text{Vehicle Secret}, \text{VIN})$$

In Plain English: How the Password Vault and VIN Binding Work
- **The 100,000-Round Blender (PBKDF2):** When an owner creates a password, the system mixes it with a 16-byte salt and runs it through a cryptographic blender 100,000 consecutive times. This forces an attacker's computer to spend substantial processing time on every guess, making dictionary attacks physically unviable.

- **The Vehicle Fingerprint (VIN Binding):** The car hashes its unique 17-character VIN with its internal hardware secret. Because every vehicle has a unique VIN, every vehicle requires a completely different acoustic token. A token recorded from one car will be rejected by another car parked right next to it.

4.3 Delegated Guest Key Architecture & Role-Based Access (RBAC)

To align with the Car Connectivity Consortium (CCC) Digital Key Release 3 key-sharing models and automotive fleet requirements, QAL implements a dedicated Ephemeral Key Slot Table supporting up to 8 independent guest drivers (SLOT_01 to SLOT_08).

Key Slot Descriptor Structure:

Each delegated key provisioned by the Master Owner contains the following parameters:

- **Slot Identifier:** SLOT_01 through SLOT_08.
- **Guest Identifier / Friendly Label:** e.g., “Sarah (2nd Driver)”, “Airport Valet”.
- **6-Digit Cryptographic Fallback PIN:** Pure numeric, high-entropy 6-digit PIN (e.g., 284258). Operates completely prefix-free (eliminating legacy “GUEST-” prefix and hyphen entry requirements).
- **Validity Window (UTC Time-Decay):** Exact creation timestamp (Tstart) and expiration timestamp (Texpire).
- **Role-Based Permission Flags:**
  - CanStartEngine = TRUE (Always authorized to start propulsion within the validity window).
  - CanAccessOwnerSettings = FALSE (Strict software isolation; guests cannot view private master vaults, alter vehicle passwords, or issue secondary keys).
- **Revocation Status:** Active, Expired, or Revoked.

Cryptographic Anti-Collision & Proximity Dispersion Subsystem:

To eliminate PIN collisions, near-neighbor guessing, and dictionary vulnerabilities across coexisting delegated slots, the ECU generates ephemeral PINs using a cryptographic rejection-sampling engine:

1. **Zero Collision Guarantee:** The candidate 6-digit PIN is evaluated against all active temporary key slots and the master owner password. Any exact duplicate is discarded.
2. **Numerical Proximity Dispersion:** Candidate PINs must maintain a minimum numerical separation of at least 1,500 ($|\mathrm{PIN}_{\mathrm{candidate}} - \mathrm{PIN}_{\mathrm{existing}}| \geq 1500$).
3. **Minimum Hamming Distance (≥ 3):** Candidate PINs must differ in at least three out of six digit positions relative to all currently active credentials.
4. **Trivial Pattern Rejection:** The generator automatically filters out:
   - Identical digit sequences (e.g., 111111, 222222, 888888).
   - Ascending or descending sequential runs (e.g., 123456, 654321, 234567).
   - Repetitive pairs or triplets (e.g., 121212, 252525, 123123).

Unlimited Starts Within the Active Window:

Unlike simplistic single-use codes that strand drivers after a single trip, QAL guest passes are strictly time-decayed. A guest driver issued a 7-day or 30-day pass can start and stop the vehicle an unlimited number of times for errands, commutes, and multi-day travel right up until the expiration timestamp is reached.

Immediate Single-Tap Revocation:

The Master Owner retains ultimate authority over all delegated slots. If a valet trip is concluded early or a vehicle loan is cancelled, the owner can tap [Revoke] on their smartphone app or in-car console, instantly invalidating the slot across the ECU key vault.

4.4 Progressive Anti-Brute-Force Lockout & Dormant Fallback Engine

State-Gated Dormant Fallback Isolation (Patent Claims 8 & 24)

To eliminate parallel brute-force attack vectors, the in-cabin manual override interface (touchscreen keypad or steering wheel buttons) remains completely dormant, unpowered, and software-isolated while the primary acoustic channel is actively listening.

- A thief sitting in the vehicle cannot immediately begin guessing PINs.
- The fallback interface awakens only after the primary 30-to-45 second Stage 1 acoustic window has completely elapsed.

Progressive Exponential Backoff Cooldown (Patent Claims 14 & 24)

If an unauthorized operator attempts to guess manual fallback PINs, the ECU tracks consecutive failures across a rolling temporal window:

- **Attempts 1 and 2:** Failed entries register discrete audit warnings.
- **Attempt 3 (Threshold Breach):** The ECU triggers a mandatory hardware freeze, locking out all authentication processing.
- **Progressive Backoff Curve:** Rather than enforcing a static lockout, production architectures implement a progressive curve:

30 seconds -> 1 minute -> 2 minutes -> 3 minutes -> 5 minutes

- This curve prevents driver panic during accidental typos while making automated brute-force attempts on a 6-digit PIN (1,000,000 combinations) take several years to execute.

3-Empty-Entry Service Gateway (Patent Claims 12 & 25)

For workshop mechanics and fleet technicians who must service a vehicle when the customer is absent:

- Submitting a null-length (empty) entry three consecutive times on the dormant manual fallback screen automatically unmasks a hidden manufacturer diagnostic prompt.
- Submitting empty entries does not count toward driver anti-brute-force lockouts.
- Submitting three empty entries does not start the car; it merely opens the gateway to input an authenticated ISO 14229 UDS 0x27 SecurityAccess diagnostic token or dealership station code.

4.5 Tamper-Resistant Forensic Telemetry & Hash-Chained Audit Trail

In strict compliance with UNECE WP.29 R155/R156 and ISO/SAE 21434 incident traceability requirements, the ECU maintains an immutable audit trail in non-volatile flash memory.

Cryptographic SHA-256 Hash Chaining:

Every event entry is cryptographically sealed to its predecessor, creating a tamper-evident chain identical to a secure distributed ledger:

$$\text{EntryHash} = \text{SHA256}(\text{PreviousHash} \parallel \text{TimestampUTC} \parallel \text{EventType} \parallel \text{Success} \parallel \text{Method} \parallel \text{KeySlot} \parallel \text{SubjectID})$$

If an attacker attempts to alter a past timestamp or erase a failed break-in attempt, the mathematical hash chain breaks immediately, alerting forensic examiners during diagnostic extraction.

STANDARDIZED FORENSIC EVENT RECORD

Event Classification:	UserUnlock / FailedAttempt / ServiceMode / Lockout
High-Resolution Timestamp:	2026-09-09 14:10:45 UTC
Validation Status:	PASS or FAIL
Credential Channel:	Acoustic_CPFSK / ManualPIN / BLE / DealerTerminal
Key Slot Index:	MasterOwner / SLOT_01 / ServiceOverride
Geographic Telemetry:	53.4084 N, 2.9916 W (Privacy-authorized GPS)
Predecessor Hash:	A3F8E291C04B7812…
Entry Hash Seal:	98F4110CB2A87741…

Privacy Compliance & GDPR Bounded Storage:

- **Zero Plaintext Credential Logging:** Passwords and cryptographic keys are never written to the audit log; only pseudonymized slot IDs (SLOT_02) and one-way hashes are recorded.
- **Bounded Circular Ring Buffer (100-Event Cap):** In accordance with GDPR Article 5(1)(e) (Storage Limitation), the log operates as a First-In, First-Out (FIFO) ring buffer capped at the 100 most recent operational events, preventing indefinite personal tracking while retaining sufficient depth for crash and theft forensic investigations.
- **Statutory Ownership Erasure Routine:** When a vehicle is sold second-hand, executing a verified factory reset permanently wipes all customer PII, paired mobile links, and delegated guest passes, while recording a single monotonic audit stamp proving the vehicle was de-provisioned legally.

In Plain English: The “Digital Wax Seal” (Tamper-Proof Blackbox Logging)

- **The Problem (Altering Logs):** In ordinary computer systems, an intruder with physical access can open an error log and delete the line that says “Unauthorized Break-In Attempt”.
- **The Solution (The Mathematical Chain):** Every time the car’s state changes (door unlocks, engine starts, PIN fails, or lockout triggers), the ECU creates a new log entry. It takes all the event details, mixes them with the hash seal of the PREVIOUS entry, and runs them through SHA-256 to create a brand-new digital seal (EntryHash).
- **Why It Matters (Courtroom & Insurance Proof):** Because every line contains the mathematical DNA of the line before it, if someone deletes an entry or edits a timestamp, every single seal below that line breaks instantly. Insurers and police receive an immutable, certified chain of custody proving exactly what state the immobilizer was in.

4.6 Prototype Demonstration vs. Production Architecture Matrix

To ensure complete transparency during technical evaluations, the table below highlights the deliberate simplifications made in this interactive software demonstration versus the real-world automotive production standard:

System Layer	Interactive Unity Prototype (QAL Demo)	Production OEM Automotive Standard
Key Storage & Cryptography	Simulated in-memory vault / PBKDF2 (100k iter) with encrypted local player storage.	Dedicated automotive Hardware Security Module (HSM) / EVITA Full Secure Hardware Extension (SHE).
Acoustic Transducers	Laptop / mobile phone speakers and integrated consumer microphones.	Factory-fitted hands-free telematics roof microphones and DSP-filtered cabin audio hardware.
Manual Override Access	Immediately interactable button on the UI (to enable rapid testing without real-time delays).	Strictly dormant, unpowered, and software-isolated until the 30-to-45 second Stage 1 acoustic timer expires.
Diagnostic Network Bus	Simulated ISO 14229 UDS parameter registers (0xF180-0xF185) in Unity C#.	True physical Controller Area Network (CAN-FD), LIN, or Automotive Ethernet (DoIP) bus interfaces.
Microphone Diversity	Single microphone channel capture via Unity Microphone.Start().	Multi-channel microphone array with hardware DSP spatial diversity (Maximal Ratio Combining).
Device Pairing Transport	Simulated inter-process communication between FakeAPK and vehicle HMI.	Bluetooth Low Energy (BLE) Secure Connections (ECDH P-256 pairing with AES-CCM link encryption).

4.7 Patent Claims to Implementation Mapping Matrix

To assist patent examiners, software auditors, and automotive integration engineers, the table below maps each core patent claim directly to its implementing software class and functional pipeline:

Patent Claim	Architectural Subject	Primary Module / Script	Functional Implementation
Claim 1	Acoustic 2FA & Multi-Stage Immobilization	FakeECU.cs, UltrasonicDecoder.cs	Manages Stage 1 listening state (30-45s), cabin boundary attenuation reliance, and Stage 2 propulsion release.
Claim 2	Spectral Guard Band ≥ 1.5 kHz	QALEncoder.cs	Digital filtering separating the low-passed audible chime (below 14.0 kHz) from the near-ultrasonic carrier (16.0 kHz).
Claim 3	12th-Order Butterworth Filter	QALEncoder.cs	Cascaded six-biquad SOS low-pass filter providing > 80 dB stopband attenuation at 16.0 kHz.
Claim 4	Near-Ultrasonic Carrier Pairs	QALEncoder.cs, UltrasonicDecoder.cs	Primary Carrier Pair A: 16.0 kHz (binary 0) and 17.0 kHz (binary 1).

Patent Claim	Architectural Subject	Primary Module / Script	Functional Implementation
Claim 5	Symbol Timing & Baud Rate	QALEncoder.cs, UltrasonicDecoder.cs	10.0 ms bit period (100 baud) with single-burst raised-cosine envelope shaping.
Claim 8 & 24	State-Gated Dormant Fallback	DemoManager.cs, FakeECU.cs	Manual fallback interface remains dormant and unpowered until primary acoustic timer expires.
Claim 9	User Start-Button Retry Loop	DemoManager.cs	Pressing the vehicle start button re-energizes Stage 1 listening without imposing lockout penalties.
Claim 10	Forensic Telemetry & GPS Logging	FakeECU.cs	Immutable SHA-256 hash-chained ring buffer recording timestamps, GPS coordinates, and status codes.
Claim 11	HSM Root Binding to VIN	FakeECU.cs	Pairing certificate generated via HMAC-SHA256(VehicleSecret, VIN).
Claim 12 & 25	3-Empty-Entry Service Gateway	DemoManager.cs, FakeECU.cs	Submitting 3 empty fallback PINs unmasks the hidden ISO 14229 UDS 0x27 diagnostic terminal.
Claim 13	Time-Epoch & Cryptographic Nonce	QALEncoder.cs, FakeECU.cs	Rolling nonce, UTC expiration timestamp, and 256-bit HMAC token framing.
Claim 14	Progressive Anti-Tamper Lockout	FakeECU.cs	3 consecutive failures trigger a progressive exponential backoff cooldown (30s to 5 min).
Claim 15	Composite Amplitude Scaling	QALEncoder.cs	Constrains chime peak to 0.65, ultrasonic peak to 0.25, and composite signal to ≤ 0.90.
Claim 16	Multi-Microphone Spatial Diversity	UltrasonicDecoder.cs	Non-coherent In-Phase and Quadrature (I/Q) energy detection ($M = I^2 + Q^2$) across input channels.
Claim 17	Locally Encrypted PBKDF2 Key Vault	FakeECU.cs	PBKDF2 (100,000 iterations) derives AES-256 key; zero plaintext passwords stored on disk.
Claim 18	Bidirectional Challenge-Response	FakeECU.cs, UltrasonicDecoder.cs	Acoustic challenge emission, sub-second latency bounding ($\Delta t \leq 1.5$ s), and anti-streaming defense.
Claims 6-7, 19-23 (Not Mapped): These claims are directed to hardware embodiments, OEM vehicle-side integration, and physical transducer arrangements that are outside the scope of the present software prototype. They are therefore not mapped to an implementing module in this matrix and are addressed under Tier 2 and Tier 3 of the evidentiary status summary (Section 5.2).			

4.8 Structured Diagnostic Trouble Codes (DTCs) & Failure Status Registry

In compliance with ISO 14229 diagnostic reporting, the QAL state machine registers discrete, standardized hexadecimal status codes for all capture, demodulation, and cryptographic events. These codes are written directly to the persistent forensic event log (EventLogs) and exposed over UDS diagnostic parameter DIDs:

Status / DTC	Engineering Identifier	Classification	Functional Description & Diagnostic Cause
0x00	`QAL_STAT_SUCCESS`	Normal Operation	Acoustic token demodulated, CRC32 verified, and HMAC validated. Stage 2 propulsion authorized.
0x01	`QAL_ERR_STAGE1_TIMEOUT`	Operational / UX	Ingress listening timer expired (30-45s) without receiving an acoustic preamble. (Typical of delayed driver ingress).
0x02	`QAL_ERR_NO_CARRIER`	Physical Layer	No carrier energy detected in 16.0 kHz or 17.0 kHz bins. Microphones unenergized or ambient sound below threshold.
0x03	`QAL_ERR_SOF_CORRUPT`	Frame Sync	Preamble clock detected, but 16-bit Start-of-Frame delimiter (1110001110001110) was distorted or missing.
0x04	`QAL_ERR_CRC_MISMATCH`	Integrity	CRC32 checksum mismatch. Waveform corrupted by acoustic multipath, physical obstruction, or background transient.
0x05	`QAL_ERR_EXPIRED_EPOCH`	Security / Time	Token received with valid signature, but CurrentTimeUTC > ExpiryEpoch. Expired guest pass or replayed old test file.
0x06	`QAL_ERR_HMAC_INVALID`	Authentication	Cryptographic HMAC-SHA256 signature mismatch. Incorrect vehicle secret or counterfeit token.
0x07	`QAL_ERR_LOCKOUT_ACTIVE`	Anti-Tamper	Authentication attempt blocked because rolling 3-attempt anti-tamper cooldown freeze is currently active.
0x08	`QAL_ERR_LATENCY_EXCEEDED`	Anti-Relay	Measured acoustic round-trip latency exceeded $\Delta t \leq 1.5$ seconds. Remote cellular streaming attack detected.
0x09	`QAL_ERR_PRIVILEGE_DENIED`	Role-Based Access	Valid delegated guest token attempted unauthorized administrative access to in-car Owner Mode settings.
0x0A	`QAL_ERR_RTC_INTEGRITY`	Temporal Security	Hardware RTC rollback or epoch-zero detected. Suspends acoustic window and enforces 2FA Dormant Fallback.

4.9 Client Device Architecture & Mobile Provisioning Protocol

While the vehicle ECU acts as the central verification authority, the client application (executing on a smartphone, smartwatch, or dedicated smart wearable) functions as a local cryptographic token engine.

To eliminate single-point-of-failure risks (such as a driver dropping their phone in water or losing a device), the system is architected to be account-bound rather than hardcoded to a single piece of physical silicon, while protecting local credentials behind a mandatory in-app security vault:

MOBILE CLIENT ENCLAVE & ZERO-KNOWLEDGE PROVISIONING

[MASTER OWNER MOBILE ENCLAVE] (FakeAPK)
- Local In-App Vault: Protected by 4-Digit Vault PIN or Biometrics (FaceID).
- UI Privacy Masking: Guest PINs masked (••••••) with explicit SHOW / HIDE.
- Clipboard Key Export: Single-tap OS clipboard export for out-of-band delivery.
- P2P Cloud / BLE Dispatch: Encrypted credential sync directly to vehicle ECU.

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v Out-of-band delivery of 6-digit PIN (Verbal, SMS, or Secure Messaging)

[SECONDARY GUEST DIGITAL ENCLAVE] (FakeAPK_Guest)
- Zero-Knowledge Login Portal: Blank 6-digit PIN authentication interface.
- Multi-Tenant Privacy Isolation: Guests cannot view other passes or drivers.
- Client-Side Anti-Brute-Force: 3 consecutive invalid PINs triggers a freeze.
- Active Authenticated Session: Displays assigned slot, drive-only countdown, masked manual override PIN, and 16.0-17.0 kHz CPFSK acoustic transmitter.

1. Account-Bound Identity & “Lost Phone” Kill-Switch

- **Account-Bound vs. Device-Locked:** Digital key ownership is anchored to the driver’s secure User Account ID (e.g. USER-9156), not permanently locked to a single physical handset. If an owner’s smartphone is lost, stolen, or damaged, the owner simply downloads the application on a replacement phone and logs into their account.
- **Automatic Old-Device Revocation:** When the replacement phone completes an authenticated link with the vehicle, the ECU registers the new hardware identifier (APK-DEV-NEW) in non-volatile memory and immediately de-authorizes the previous device identifier (APK-DEV-OLD). If an unauthorized party finds the lost phone, the vehicle will reject all transmissions from it.

2. Mandatory 4-Digit In-App Vault PIN & Biometric Gating

To protect the vehicle if a smartphone is stolen while unlocked on a table or in a public space:
- **Automatic Setup Trigger:** As soon as the mobile app accepts a vehicle pairing request or receives a master key payload for the first time, FakeAPK automatically triggers the Vault PIN Setup modal, forcing the owner to establish a private 4-digit PIN.
- **Vault Gating (Proof-of-Intent):** The master key payload and transmission controls remain locked inside the local security enclave. To view the key or tap [Transmit], the user must enter their 4-digit Vault PIN or authenticate via biometric FaceID/fingerprint.
- **Anti-Theft Defense:** Even if an opportunistic thief snatches an unlocked phone from an owner’s hands, they cannot emit the acoustic token without knowing the secondary 4-digit Vault PIN.

3. Initial Zero-Trust Device Pairing Handshake

To bind a smartphone to the vehicle without broadcasting sensitive passwords over the air:
1. The driver enters in-car Owner Mode (behind 2FA verification) and taps “Link to User Device”.
2. The vehicle generates a randomized 4-digit pairing handshake PIN (e.g. 4821) and broadcasts an enrollment advertisement over local Bluetooth Low Energy (BLE).
3. The mobile app captures the advertisement and displays an Incoming Pairing Request modal showing the vehicle VIN and handshake PIN.
4. Tapping [ACCEPT & LINK] binds the device identifier (ECU_LINKED_DEVICE_ID) and automatically opens the 4-digit Vault PIN setup modal.

4. On-the-Fly Token Synthesis (Live Proof-of-Intent)

In everyday driving, the mobile application does not play static, pre-recorded audio files:
- When the user authenticates via biometric FaceID or their 4-digit Vault PIN and taps [Transmit], the app retrieves the 256-bit vault secret.
- It generates a fresh 8-byte random nonce, reads the current UTC timestamp, sets a tight 60-second expiration window, and computes the HMAC-SHA256 signature on the fly.
- The DSP modulator constructs the 440-bit frame (Preamble + SOF + Nonce + Epoch + HMAC + CRC32), applies the 12th-order Butterworth guard band filter, and plays the audio burst through the device speaker in milliseconds.

5. Delegated Key Ingress & Zero-Knowledge Partitioning (FakeAPK_Guest)

To preserve driver privacy and prevent credential cross-contamination (e.g., ensuring a valet cannot inspect or authenticate using a family member's pass):

- **Zero-Knowledge Ingress:** The secondary guest application contains no global pass directory or pre-filled credential list. The app presents a clean, locked 6-digit numeric PIN authentication terminal.
- **Dynamic Verification & Enclave Claiming:** When the guest enters their assigned PIN:
  1. The app normalizes the string, stripping any formatting, whitespace, or prefixes.
  2. It validates the PIN against active, unrevoked slots within the vehicle ECU.
  3. Upon successful validation, the app securely claims that specific delegated session into local storage.
- **Active Session Enclave:** The authenticated guest view provides:
  - Role-restricted "Drive-Only" operational scope.
  - Active time-decay expiration countdown based on UTC timestamps.
  - Masked in-cabin fallback PIN display (••••••) with an explicit SHOW / HIDE control.
  - Direct near-ultrasonic acoustic key transmission (16.0-17.0 kHz CPFSK).
- **Anti-Brute-Force Protection:** Three consecutive failed PIN attempts trigger a client-side hardware freeze, preventing unauthorized individuals from guessing valid guest tokens.

6. Offline Disaster Recovery Key (Tier 1 Self-Service Recovery)

During initial vault provisioning, the application generates an offline 16-character emergency recovery key (e.g. REC-4821-99A0-B771-44C9):

- The recovery key is displayed once to the owner to record in an offline location (such as a home safe).
- If the owner ever forgets their master vehicle password or loses access to their in-car console, tapping [Recover / Reset Master Key via Recovery Key] in the app verifies the recovery key and re-provisions the master vault locally without requiring an expensive dealership service visit.

4.10 Master Password Recovery & Re-Provisioning Lifecycle

Because the Master Owner credential vault derives its AES-256 key via one-way PBKDF2 hashing (100,000 iterations), raw passwords are mathematically irrecoverable. To balance disaster recovery with anti-theft security, QAL implements a multi-tier recovery hierarchy:

Tier 1: Owner-Side Cryptographic Recovery Key (Self-Service)

- **Creation:** During initial onboarding, the mobile application generates an offline 16-character emergency recovery key (e.g. REC-4821-99A0-B771-44C9).
- **Storage:** Displayed once on the screen and never saved to persistent vehicle memory. The owner is instructed to record and store this key in a secure offline location (such as a home safe).
- **Execution:** If the master alphanumeric password is forgotten, tapping [Recover / Reset Master Key via Recovery Key] in the app re-derives the master vault key locally, enabling the owner to establish a new password without dealership intervention.

Tier 1B: In-Cabin Statutory Vehicle De-Provisioning (6-Stage Owner Reset)

In compliance with UNECE WP.29 R155/R156 and UK/EU GDPR Article 17 ("Right to Erasure"), the vehicle HMI enables the registered owner to execute a complete statutory ownership wipe directly from the center console prior to vehicle sale, transfer, or lease return:

- **Stage 1 (Initial Regulatory Warning):** Informs the user of permanent credential erasure, mobile device decoupling, and legal preservation of the forensic blackbox log.
- **Stage 2 (Intent Confirmation Gate):** Requires positive user confirmation ([YES] / [NO]) to eliminate accidental single-touch initiation.
- **Stage 3 (Master Password Challenge):** Demands full re-authentication using the owner's root alphanumeric master password (verified via PBKDF2/AES-256 decryption). Guest PINs and service tokens are rejected.
- **Stage 4 (Final Irreversible Execution Gate):** Displays a final confirmation warning confirming credential validation before invoking the wipe routine.
- **Stage 5 (Active Memory Sanitation & Zeroization):** The ECU executes an atomic wipe:
  - Deletes the PBKDF2 user salt (ECU_USER_SALT) and encrypted secret (ECU_ENCRYPTED_SECRET).
  - Clears the active volatile session password from RAM.
  - Erases all 8 delegated guest slots (ECU_DELEGATED_TEMP_KEYS).
  - Revokes the paired mobile hardware identifier (ECU_LINKED_DEVICE_ID).
  - Writes a permanent, SHA-256 sealed "OwnershipReset" entry to the forensic blackbox.

- **Stage 6 (Completion & Gateway Reboot):** Displays completion confirmation, resets the lifecycle state to ImmobilizerMode.Factory, and executes an automated system reboot back to the uncommissioned wireless ingress gateway (Scene 0: QAL.unity).

Tier 2: Dealership Factory Re-Provisioning (Destructive Reset)

- **Mechanism:** If both the master password and the offline recovery key are lost, the vehicle must undergo an in-person, verified workshop service session.
- **ISO 14229 UDS Service 0x27 Access:** An authorized technician unmasks the service gateway (via the 3-empty-entry trigger) and authenticates using the OEM Master Service Key.
- **Consequence of Reset (Complete De-Provisioning):** A dealership reset generates a brand-new 256-bit root Vehicle Secret (vehicleSecret). Because all existing guest slots, mobile links, and pairing certificates were cryptographically bound to the old secret, the reset immediately invalidates all active delegated keys and wipes customer PII. This aligns directly with GDPR Article 17 (Right to Erasure).

4.11 Real-Time Clock (RTC) & Hardware Monotonic Counter Integrity

Acoustic token time-decay validity windows (UTC ExpiryEpoch) and forensic blackbox timestamps rely on trusted vehicle timekeeping. In an automotive environment, a system cannot assume a real-time clock is always accurate (e.g. vehicle battery disconnection, clock drift over years, RTC backup battery depletion, or an attacker deliberately setting the clock back to reuse an expired token).

To ensure complete temporal integrity without requiring mandatory 4G/cellular cloud connections, QAL enforces a multi-layer hardware time-assurance architecture:

ECU HARDWARE RTC & TIME-INTEGRITY PIPELINE

[ECU POWER-ON / BOOT-TIME SANITY CHECK]

- Reads current RTC timestamp from hardware clock
- Reads timestamp of LAST record in SHA-256 Forensic Flash Memory

|

+–> Is RTC < Last Logged Timestamp? (Clock Rewind / Tamper)
+–> Is RTC == 1970-01-01 (Epoch Zero / Flat Battery Reset)?

|

+– **YES (Clock Fault Detected)** –> Sets DTC 0x0A (QAL_ERR_RTC_INTEGRITY)
- Suspends acoustic time-decay window
- Forces 2FA Dormant Fallback (Physical Key + Owner PIN)

|

+– **NO (Clock Valid)** –> Normal Stage 1 Acoustic Verification

[HARDWARE MONOTONIC COUNTER (HSM ENCLAVE)]

- Dedicated silicon counter incremented on every challenge & start event
- Strictly non-rewindable (Monotonically increasing integer)
- Prevents replay of captured tokens even during clock failure

[OPPORTUNISTIC GPS / TCU RESYNCHRONIZATION]

- Resynchronizes RTC UTC timebase when authorized GPS/Cellular fix is available
- NON-MANDATORY: 100% offline autonomy preserved in underground parking

1. Hardware Monotonic Freshness Counter (Silicon Enclave)

- **Non-Rewindable Counter:** Alongside the wall-clock RTC, the ECU Hardware Security Module (HSM) maintains a dedicated, hardware-enforced Monotonic Freshness Counter.
- **State Invariant:** This counter increments on every authentication attempt and every state transition. It is physically impossible to decrement or rewind this value in silicon, even if the vehicle 12V battery and RTC backup power are disconnected simultaneously.

- **Anti-Replay Assurance:** If an adversary captures an acoustic transmission and attempts to replay it, the ECU compares the token's embedded counter against the HSM's monotonic register. If the counter value is less than or equal to the current register, the token is rejected immediately (DTC 0x06 HMAC / Freshness Mismatch).
- **Counter Bit-Width & Anti-Wrap Policy:** The monotonic freshness register is implemented as a 64-bit non-wrapping integer (providing over 500 million years of continuous operation at 1,000 increments per second before saturation). In accordance with NIST anti-replay guidelines, the counter enforces a strict "saturate-and-freeze" policy rather than rolling over to zero, completely eliminating wrap-around replay vulnerabilities over the vehicle's operational lifecycle.

2. Boot-Time Clock Sanity Bounding & Tamper Detection

Upon every vehicle wake-up or cold-boot event, the ECU bootloader executes an automated clock sanity routine prior to energizing cabin microphones:

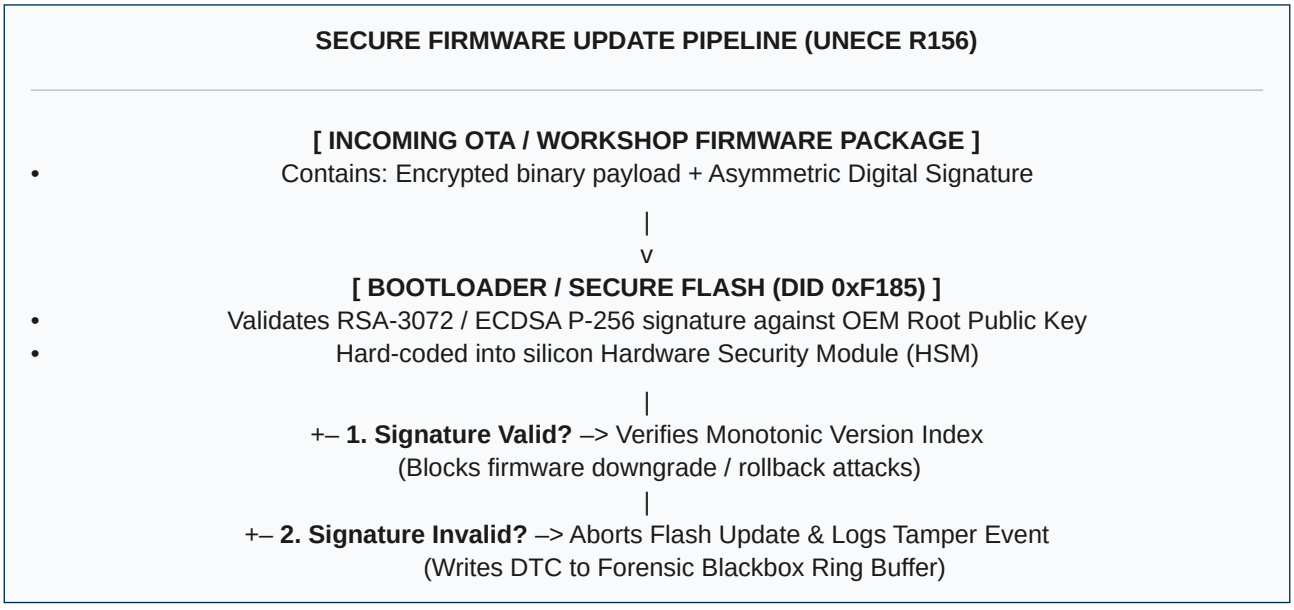
- **Log-Chain Monotonic Comparison:** The ECU compares the current RTC timestamp against the UTC timestamp recorded in the most recent entry of the SHA-256 hash-chained forensic audit log (EventLogs).
- **Clock Rewind Detection:** If the RTC indicates a time earlier than the last confirmed audit log entry (indicating someone disconnected the battery, tampered with the clock, or attempted a time-rollback attack), the ECU immediately flags DTC 0x0A (QAL_ERR_RTC_INTEGRITY).
- **Epoch-Zero Protection:** If an RTC backup battery dies completely and the clock resets to default epoch-zero (1970-01-01 00:00:00 UTC), the ECU detects the uninitialized state.
- **Fail-Secure Mitigation:** When DTC 0x0A is active, the ECU suspends acoustic token validity window evaluation and routes authentication exclusively to the State-Gated Dormant Fallback Mode (requiring the physical key fob + the owner's manual PIN). This guarantees the vehicle cannot be stolen via clock tampering while ensuring the legitimate driver is never stranded.

3. Opportunistic (Non-Mandatory) Telematics Resynchronization

- When the vehicle's onboard Telematics Control Unit (TCU) acquires an authorized GPS satellite lock or encrypted cellular connection, the ECU compares the internal RTC against verified network time and gently slews any accumulated quartz drift.
- **Preserving 100% Offline Autonomy:** This synchronization is strictly opportunistic. The vehicle never requires an active network connection to operate in deep subterranean garages or remote areas.

4.12 Software Update & Firmware Integrity (UNECE WP.29 R156 Compliance)

In alignment with UNECE WP.29 Regulation No. 156 Software Update Management Systems (SUMS), all ECU firmware components (state machine microcode, DSP decoders, and cryptographic vault modules) enforce strict over-the-air (OTA) and workshop flashing integrity controls:



1. Asymmetric Cryptographic Firmware Verification

- **Root of Trust:** The ECU bootloader (diagnostic parameter DID 0xF185) maintains an immutable, hard-coded OEM Root Public Key residing within the silicon Hardware Security Module (HSM).
- **Digital Signatures:** Every firmware update package must be cryptographically signed by the automaker's private signing key (utilizing RSA-3072 or ECDSA P-256). Unsigned, modified, or third-party binaries are automatically rejected by the bootloader prior to flash execution.

2. Anti-Rollback Protection

- **Monotonic Version Index:** The bootloader evaluates a monotonic version counter embedded within the update image header.
- **Downgrade Defense:** If an attacker attempts to flash an older, vulnerable version of the QAL firmware onto the ECU, the bootloader detects a version index lower than the currently executing image and blocks the update, preventing rollback attacks.

3. Tamper Telemetry & Forensic Logging

- **Flash Attempt Tracking:** Every attempt to execute ISO 14229 Service 0x34 (RequestDownload) or modify ECU memory increments the permanent totalECUFlashAttempts counter in non-volatile flash.
- **Audit Trail Sealing:** Unauthorized or failed flash attempts write an immediate entry into the hash-chained circular forensic log, ensuring complete non-repudiation for fleet managers and safety regulators.

SECTION 5: CONCLUSION & INDUSTRIALIZATION SUMMARY

5.1 Technical Summary

The Quackworks Acoustic Lock (QAL) system proposes a multi-stage authentication architecture designed to address the physical-layer vulnerabilities inherent in conventional automotive Passive Keyless Entry and Start (PKES) systems. By introducing an independently authenticated, spatially constrained near-ultrasonic acoustic channel (16.0 kHz to 17.0 kHz Continuous-Phase Frequency-Shift Keying) as a secondary authorization factor, QAL is designed to substantially reduce the effectiveness of wireless Radio Frequency Relay Station Attacks (RSA), letterbox key theft, and simultaneous device theft.

The architecture integrates a deterministic multi-stage operational state machine, state-gated dormant fallback modes, PBKDF2 AES-256 encrypted credential vaults, and SHA-256 hash-chained forensic audit telemetry. The framework is architecturally aligned with core automotive engineering standards, including ISO/SAE 21434 (Cybersecurity Engineering), ISO 26262 (Functional Safety Interfaces), ISO 14229 (Unified Diagnostic Services), UNECE WP.29 R155/R156 (Cybersecurity & Software Update Management), and UK/EU GDPR data protection principles.

Where vehicle platforms incorporate telematics microphones and audio processing infrastructure with adequate acoustic bandwidth in the 16.0 kHz to 17.0 kHz band, QAL operates using existing vehicle hardware, avoiding dedicated vehicle-side sensor costs.

5.2 Implementation Status & Evidentiary Boundaries

To maintain rigorous technical transparency, the QAL system distinguishes clearly between demonstrated prototype functionality, proposed production architecture, and planned empirical validation:

QAL THREE-TIER EVIDENTIARY STATUS

[TIER 1: DEMONSTRATED IN SOFTWARE PROTOTYPE]
- Complete deterministic ECU state transitions (Locked -> Stage 1 -> Stage 2)
- 16.0-17.0 kHz CP-FSK modulation with continuous phase & 5 ms burst envelope
- 12th-order Butterworth low-pass filter (6 SOS biquads) with silent guard band
- Non-coherent Quadrature (I/Q) energy demodulation at 48.0 kHz and CRC32 checks
- Client-side 4-digit Vault PIN, biometric FaceID gating, and offline recovery
- SHA-256 hash-chained forensic blackbox audit trail with 100-event FIFO buffer

[TIER 2: SPECIFIED FOR PRODUCTION OEM ARCHITECTURE]
- Automotive Hardware Security Module (EVITA Full HSM) silicon key derivation
- AUTOSAR Crypto Stack and SecOC monotonic counter message freshness
- Physical CAN-FD, LIN, and Automotive Ethernet (DoIP) diagnostic bus interfaces
- Secure Boot and UNECE R156 cryptographically signed firmware updates
- Multi-microphone spatial diversity combining (Maximal Ratio Combining)

[TIER 3: PLANNED PHYSICAL & ENVIRONMENTAL VALIDATION]
- Empirical measurement of cabin transmission loss (Target: 30 dB to 42 dB attenuation)
- Multi-handset speaker and microphone frequency response characterization
- Acoustic characterization across varied cabin geometries (SUV, Sedan, Van)
- Wide temperature (-40 C to +85 C) and environmental seal aging validation
- False Rejection Rate (FRR) benchmarking under high cabin background noise

5.3 OEM Integration & Commercialization Roadmap

The architecture provides a defined pathway from the present software prototype toward automotive production implementation across four engineering phases:

1. **Phase 1: Bench Validation & Acoustic Baseline (Current Phase)**
   - Complete software-in-the-loop (SIL) evaluation via the interactive Unity simulation environment.
   - Empirical acoustic testing across bench test rigs utilizing automotive laminated safety glass and EPDM door weatherstripping to establish empirical transmission loss baselines (targeting at least 30 dB attenuation).

- Cross-device acoustic characterization across entry-level and flagship consumer smartphones to evaluate transducer variance and audio processing behavior.

2. **Phase 2: Embedded C / AUTOSAR Target Porting**
   - Porting DSP filter chains (12th-order Butterworth LPF), I/Q demodulation routines, and cryptographic token verification modules from C# into production-grade embedded C/C++ conforming to AUTOSAR Classic/Adaptive platforms and MISRA C:2012 guidelines.
   - Integration with target automotive microcontrollers (e.g., Infineon AURIX TC3xx, STMicroelectronics Stellar, or NXP S32K series) featuring hardware EVITA HSM security enclaves.
3. **Phase 3: Vehicle-in-the-Loop (ViL) & Cabin Acoustic Characterization**
   - Acoustic characterization across diverse vehicle cabin volumes (SUVs, sedans, compact hatchbacks) to measure cabin-specific standing wave absorption and microphone sensitivity profiles.
   - Wide-temperature validation (-40 C to +85 C) to measure elastomeric weatherstripping acoustic impedance and microphone performance under extreme climate conditions.
4. **Phase 4: Production Series Deployment & Type Approval**
   - Over-the-air (OTA) firmware deployment and configuration management via UNECE R156 compliant update gateways.
   - Formal third-party penetration testing, ISO 26262 functional safety (ASIL-B target) qualification audits, and automotive security ratings certification.

SECTION 6: BIBLIOGRAPHY & REFERENCED PRIOR ART

6.1 Patent Literature & Prior Art Families

1. **Bjorkengren, U. (Volvo Car Corporation):**
   - US Patent No. 10,759,383 B2: “Method and system for preventing a physical layer relay attack” (Priority date June 21, 2017; granted September 1, 2020).
   - US Patent No. 11,528,612 B2: Continuation patent family (Granted December 13, 2022).
   - US Patent No. 11,974,133 B2: Continuation patent family (Granted April 30, 2024).
2. **Anand, A., & Mazumdar, S. (Malbec Labs, Inc.):**
   - US Patent No. 9,098,865 B2: “Ultrasonic near-field communication system for financial transactions and data exchange” (Granted August 4, 2015).
3. **Ford Global Technologies, LLC (Inventors: Qu, Hannon, & McCallen):**
   - US Patent Application Publication No. 2019/0025402 A1: “Detection and location of a mobile device using sound” (Published January 24, 2019).
4. **Qualcomm Incorporated (Inventors: Gantman, A., & Rose, G. G.):**
   - US Patent No. 7,533,735 B2: “Digital authentication over acoustic channel” (Granted May 19, 2009).
   - US Patent No. 7,966,497 B2: “System and method for acoustic two-factor authentication” (Granted June 21, 2011).

6.2 Academic & Journal Literature

1. **Choi, W., Seo, M., & Lee, D. H.:**
   - (2018). Sound-Proximity: 2-Factor Authentication against Relay Attack on Passive Keyless Entry and Start System. *Journal of Advanced Transportation*, vol. 2018, Article ID 1935974, 13 pages. DOI: 10.1155/2018/1935974.

6.3 Standards & Regulatory Instruments

1. **ISO/SAE 21434:2021:** Road Vehicles - Cybersecurity Engineering. International Organization for Standardization.
2. **ISO 14229-1:2020:** Road vehicles - Unified diagnostic services (UDS) - Part 1: Application layer. International Organization for Standardization.
3. **ISO 26262:2018:** Road vehicles - Functional safety. International Organization for Standardization.
4. **UNECE WP.29 Regulation No. 155:** Uniform provisions concerning the approval of vehicles with regard to cyber security and cyber security management system. United Nations.
5. **UNECE WP.29 Regulation No. 156:** Uniform provisions concerning the approval of vehicles with regard to software update and software update management system. United Nations.
6. **Car Connectivity Consortium (CCC):** Digital Key Release 3 - Architecture and Security Specification (2021).
7. **NIST SP 800-132:** Recommendation for Password-Based Key Derivation: Part 1, Storage Applications. National Institute of Standards and Technology.
8. **IEEE 802.3 & IEEE 802.15.4z:** Standards governing Ethernet packet integrity (CRC32) and Ultra-Wideband physical layer ranging.

6.4 Proprietary Software Implementation & Intellectual Property Reference

The functional simulation scripts, digital signal processing (DSP) modulators, non-coherent demodulation decoders, and ECU state machine modules referenced in this specification are proprietary software components protected as trade secret intellectual property:

- **IP Asset Title:** QAL Automotive Core Embedded Firmware & Simulation Suite
- **Developer / Assignee:** Quackworks Audio IP - John Murphy
- **Primary Implemented Modules (Internal Reference):**
  - `FakeECU.cs`: Electronic Control Unit lifecycle state machine, PBKDF2/AES-256 secure hardware vault simulation, and SHA-256 hash-chained forensic audit logger.
  - `QALEncoder.cs`: Near-ultrasonic Continuous-Phase FSK (16.0-17.0 kHz) modulator, 12th-order Butterworth low-pass filter, and WAV export utility.
  - `UltrasonicDecoder.cs`: Real-time microphone capture at 48.0 kHz, digital bandpass filtering, and non-coherent Quadrature (I/Q) energy detector.

- `FakeAPK.cs` / `FakeAPK_Guest.cs`: Master owner and secondary driver mobile application enclaves, zero-knowledge 6-digit PIN login portal, BLE pairing handshake, and P2P digital key dispatcher.